ACTIVATED SLUDGE SYSTEM SIMULATION PROGRAMS

NITRIFICATION
AND
NITRIFICATION/DENITRIFICATION SYSTEMS

VERSION 1.0

by

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FOREWORD

Since the early 1970s significant developments have taken place in activated sludge technology, particularly in the biological nutrient (nitrogen and phosphorus) removal field. Today single sludge systems are available that will remove carbonaceous material, nitrogen and phosphorus, mediated by biological processes. Progressively, as objectives have expanded from carbonaceous material removal only, to include nitrification, denitrification and biological excess phosphorus removal, and as the effluent quality requirements have become more stringent, so activated sludge systems have become more complex, requiring multi-reactor configurations comprising aerated and unaerated reactors and inter-reactor recirculation flows. Concomitantly the number of biological processes influencing the effluent quality, and the number of compounds involved in these processes, have increased. With such complexity the performance of any proposed system can be determined only by experimentation or by a mathematical model that simulates the response accurately.

The Water Research Commission (WRC) has sponsored research into the activated sludge system at the University of Cape Town (UCT) since 1973. The research has resulted in the formulation of a comprehensive mathematical model for describing the behaviour of single sludge activated sludge systems incorporating aerated and unaerated reactors. However, the model does not yet include biological excess phosphorus removal; this aspect is being actively pursued.

The model has found application in the development of optimal design procedures, identification and solution of operational problems, and development of plant control strategies. The value of the model has been recognized by the International Association on Water Pollution Research and Control (IAWPRC); their Task Group on Modelling of Wastewater Treatment suggested some modifications to the UCT model and made a considerable contribution in the method for presenting this and other biological models.

The WRC requested the authors to develop user-friendly interactive computer programs of both the UCT and IAWPRC model versions, suitable for personal computers. By sponsoring development of the programs the Commission aims to facilitate transfer of technology to the end user, persons involved in the design, management and operation of activated sludge systems.

P E Odendaal

Executive Director
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- The numerous postgraduate students who over the years have passed through the laboratory; the model is a synthesis of their ideas and research effort.

SCOPE OF MODELS

The ***Activated Sludge System Simulation Programs – Nitrification and nitrification/denitrification systems***, includes two programs for simulation of single sludge activated sludge systems incorporating carbonaceous energy removal, nitrification and denitrification. The two programs, **UCTOLD** and **IAWPRC**, are based on what are, arguably, the two most up to date mechanistic models of the activated sludge system. **UCTOLD** is based on the model developed by the research group at the University of Cape Town (UCT) over the past fifteen years. **IAWPRC** is based on a model proposed by a Task Group of the International Association on Water Pollution Research and Control (IAWPRC) incorporating some modifications by the UCT group. With appropriate calibration the two models give predictions that are closely equal for most situations.

Both programs presented with this manual can be applied to predict steady state and cyclic dynamic response behaviour for a range of system reactor configurations, operating conditions and waste flow and loads:

- The programs can be used to predict response of single or in-series *completely mixed* reactor systems with or without inter-reactor recycles (recycles opposite to the direction of flow through the system) over the range of temperatures from 14 to 22°C. The reactors may be aerated or unaerated and the sludge age may vary from 2 to 30 days. Waste flow and/or loads may be steady state or cyclic.

- The programs predict the response of the following compounds with their various contributory components; chemical oxygen demand (COD), oxygen, volatile suspended solids (as COD), nitrogen and alkalinity.

Application of the programs has some limitations:

- The programs cannot be used to predict the behaviour of systems in which within a single reactor a part is anoxic and a part aerobic.

- The programs do not provide for prediction of biological excess phosphorus (P) removal. Prediction of P removal behaviour requires a substantial increase in the number of processes and compounds to account for the response of the P removal organisms (which can constitute up to 40 percent of the organism mass in P removal systems treating municipal effluents). Despite this constraint experience has shown that the models give reasonable prediction of the sludge masses, nitrification/denitrification, oxygen utilization and effluent COD and nitrogen of single sludge biological excess P removal systems. A separate model to include biological excess P removal is under development.

- The programs have been developed from observations made on systems treating municipal waste flows at laboratory, pilot and full scale. Default values assigned to the constants in the programs, in particular to wastewater characteristics and organism growth rates, are approximate averages observed when treating South African municipal waste flows for temperatures in the range 14 to 22°C. Application of the programs to systems treating municipal waste flows in other countries may require that some of the constants have to be redetermined - the characteristics of such waste flows may consistently differ from those in South Africa, and/or temperatures may fall outside the 14 to 22°C range. In this manual the user is given some direction as to the situations under which the constants may change. Also, procedures for independent determination of some of these constants are briefly set out. (The programs have not been tested to simulate systems treating industrial waste flows).

NOTICES

- The views expressed in this manual are those of the authors and do not necessarily constitute endorsement or recommendation by the Water Research Commission.
- The contents of this manual are subject to change without notice.
- All efforts have been made to ensure the accuracy of the programs and the contents of this manual. Should users find any errors the Water Research Commission will greatly appreciate being informed.
- The above notwithstanding, the Water Research Commission and the authors can assume no responsibility for any errors in the programs or the consequences of such errors.

- **IMPORTANT** - The computer programs presented with this user manual are a codification and rationalization of research investigation into kinetic behaviour of the activated sludge system, in particular over the last 15 years. These computer programs are very easy and simple to use by following this user manual, even for those who do not have much experience and knowledge with the activated sludge system. However, for those without considerable knowledge and experience in activated sludge behaviour, these programs should <u>not</u> be regarded as replacement for knowledge and experience. It is strongly recommended that a working knowledge and sound familiarity is held by the user on the research embodied in the models incorporated in the programs. Only with such knowledge does the user have the insight and critical ability to determine the reasonableness of the computer output for the particular application and so ensure that the programs are being appropriately used and their output meaningfully interpreted. A list of references outlining the research on which the models are based is given at the back of this user guide.

DISCLAIMER

While it is believed that the programs are based on the best available knowledge and that considerable effort has been expended in eliminating errors, users are warned that in the application of the programs the results obtained are the sole responsibility of the user.

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LIST OF SYMBOLS

The symbols for the various compounds, kinetic constants, stoichiometric constants and switching function parameters used in UCTOLD and IAWPRC are listed in this section. Details of the symbol system are given in Appendix D.

PARAMETERS FOR UCTOLD

The following compounds, kinetic constants, stoichiometric constants and switching function parameters are utilized in the model UCTOLD.

Compounds

Z_{BH} – biological (active) heterotrophic mass
Z_E – endogenous mass
Z_{BA} – biological (active) autotrophic mass
S_{ads} – adsorbed slowly biodegradable substrate
S_{enm} – enmeshed slowly biodegradable substrate
Z_I – inert mass
N_{obp} – particulate biodegradable organic nitrogen
S_{bs} – soluble readily biodegradable substrate
N_a – ammonia nitrogen
N_{obs} – soluble biodegradable organic nitrogen
N_{O3} – nitrate nitrogen
Alk – H_2CO_3* alkalinity
S_{us} – soluble unbiodegradable substrate
O – dissolved oxygen

Kinetic parameters

$\hat{\mu}_H$ – maximum specific growth rate of the heterotrophs when utilizing soluble readily biodegradable COD
K_{SH} – half-saturation coefficient for heterotroph growth when utilizing soluble readily biodegradable COD
K_{MP} – maximum specific growth rate of the heterotrophs when utilizing adsorbed particulate slowly biodegradable COD
K_{SP} – half-saturation coefficient for heterotroph growth when utilizing adsorbed particulate slowly biodegradable COD
b_H – heterotrophic organism specific death rate
K_A – slowly biodegradable COD adsorption rate
K_R – conversion rate of soluble organic nitrogen to free and saline ammonia
η_G – correction factor for anoxic heterotrophic growth

$\hat{\mu}_A$ – maximum specific growth rate of the autotrophs (nitrifiers)
K_{SA} – half-saturation coefficient for autotroph growth on ammonia
b_A – autotrophic organism specific death rate

Stoichiometric parameters

Y_{ZH} – heterotroph yield in COD units
Y_Z – autotroph yield in COD units
f_{MA} – maximum ratio of adsorbed substrate to heterotroph mass
f_E – inert fraction of the active organism mass
$f_{ZB,N}$ – fraction of biological (active) mass which is nitrogen
$f_{ZE,N}$ – fraction of endogenous/inert mass which is nitrogen

Switching function parameters

K_{OH} – switching constant for oxygen-limited heterotroph growth (switch from aerobic to anoxic growth)
K_{OA} – switching constant for oxygen-limited autotrophic growth
K_{HA} – switching constant for ammonia-limited aerobic heterotroph growth
K_{NO} – switching constant for nitrate-limited anoxic heterotroph growth

PARAMETERS FOR IAWPRC

The following compounds, kinetic constants, stoichiometric constants and switching function parameters are utilized in the model **IAWPRC**.

Compounds

Z_{BH} – biological (active) heterotrophic mass
Z_E – endogenous mass
Z_{BA} – biological (active) autotrophic biomass
S_{enm} – enmeshed slowly biodegradable substrate
Z_I – inert mass
N_{obp} – particulate biodegradable organic nitrogen
S_{bs} – soluble readily biodegradable substrate
N_a – ammonia nitrogen
N_{obs} – soluble biodegradable organic nitrogen
N_{O3} – nitrate nitrogen
Alk – H_2CO_3* alkalinity
S_{us} – soluble unbiodegradable substrate
O – dissolved oxygen

Kinetic parameters

$\hat{\mu}_H$ – maximum specific growth rate of the heterotrophs
K_{SH} – half-saturation coefficient for heterotroph growth
K_H – maximum specific hydrolysis rate
K_X – half-saturation coefficient for hydrolysis
b_H – heterotrophic organism specific death rate
K_R – conversion rate of soluble organic nitrogen to free and saline ammonia
η_S – correction factor for anoxic hydrolysis
η_G – correction factor for anoxic heterotrophic growth

$\hat{\mu}_A$ – maximum specific growth rate of the autotrophs (nitrifiers)
K_{SA} – half-saturation coefficient for autotroph growth
b_A – autotrophic specific organism death rate

Stoichiometric parameters

Y_{ZH} – heterotroph yield in COD units
Y_{ZA} – autotroph yield in COD units
f_E – inert fraction of the active organism mass
$f_{ZB,N}$ – fraction of biological (active) mass which is nitrogen
$f_{ZE,N}$ – fraction of endogenous/inert mass which is nitrogen

Switching function parameters

K_{OH} – switching constant for oxygen-limited heterotroph growth (switch from aerobic to anoxic growth)
K_{OA} – switching constant for oxygen-limited autotroph growth
K_{HA} – switching constant for ammonia-limited aerobic heterotroph growth
K_{NO} – switching constant for nitrate-limited anoxic heterotroph growth

PARAMETERS DEFINING INFLUENT WASTEWATER CHARACTERISTICS

Carbonaceous material

$f_{S,us}$	– fraction of the total influent COD which is unbiodegradable soluble
$f_{S,up}$	– fraction of the total influent COD which is unbiodegradable particulate
f_{bs}	– fraction of the biodegradable influent COD which is readily biodegradable

Nitrogenous material

$f_{N,a}$	– fraction of the total influent TKN which is free and saline ammonia
$f_{N,ous}$	– fraction of the total influent TKN which is organic unbiodegradable soluble
$f_{Nob,p}$	– fraction of the biodegradable organic TKN which is particulate

CHAPTER 1

INTRODUCTION

BACKGROUND

In the past, design, operation and control of activated sludge systems were based on relatively simplistic ideas about the behaviour of the systems, and on experience acquired in running such systems. Since about 1970 extensive development has taken place in the activated sludge method of treating wastewaters. The function of the single sludge system has expanded from carbonaceous energy removal to include nitrification, denitrification and phosphorus removal, all of these mediated biologically. These extensions have impacted on the system configuration in that multiple in-series reactors, some aerated and others not, with inter-reactor recycles have to be incorporated.

Not only have the system configuration and its operation increased in complexity, but more stringent standards for effluent quality have to be satisfied. To meet these effluent standards, the design of the selected system must be optimized and the system operated optimally. With such complexity it is no longer possible to make a reliable quantitative, or sometimes even qualitative prediction as to the effluent quality to be expected from a design, or to assess the effect of a system or operational modification without some model of the system behaviour.

To design the system optimally and to operate it effectively, concerted efforts have been made over the past 15 to 20 years to model the behaviour of these systems. This has been the case in South Africa, where development of a reliable mathematical model has been given intensive attention. The result of this research endeavour is a very powerful model that gives a reliable description of the nitrification and nitrification/denitrification (ND) system response over wide ranges of system configuration (single and in-series reactor systems, aerated and non-aerated reactors, inter-reactor recycles), wastewater characteristics (COD, TKN, flow pattern) and operational parameters (sludge age, temperature, dissolved oxygen concentration). Although this model is grounded on a mechanistic interpretation of the behaviour of the organisms mediating the process reactions, this interpretation is a subjective one, a gross simplification of the complex nature of the phenomena. However, that the model gives a reliable description of the system response over a wide range of conditions indicates that this simplification is acceptable.

The model has had a significant impact on design and operational procedures of single sludge nitrification and ND activated sludge systems. With regard to design, the model has led to the identification of procedures to estimate the optimal or near optimal design configuration, reactor sizes and operational parameters (e.g. sludge age) and to estimate the expected response (WRC, 1984). *Once the system has been designed*, the time response under dynamic flow and load conditions can be estimated using the model. Thereupon the design can be modified if necessary to achieve improved performance, or, the sensitivity of the design to changes in flow and load conditions or to operational modifications can be assessed. In full-scale plant operation it has also found application in assessing the effects of changes in waste flows and loads, operational modifications (e.g. changes in recycles), and proposed modifications to plant configuration. It has also proved valuable in operator training; through simulation exercises using the model the operator acquires 'instant' experience in the behaviour to be expected with changes in inputs, system configuration and operational strategies.

To make the model more widely available at the different levels of application, a computer program of the model is now presented that (1) is suitable for personal computers, (2) is 'user friendly', (3) is flexible, and (4) provides rapid solutions, numerical or graphical.

COMPUTER PROGRAMS

This manual accompanies two computer programs of mechanistic models for simulating activated sludge wastewater treatment system behaviour on IBM PC or compatible computers. The first program, **UCTOLD**, is based on the model developed by the research group at the University of Cape Town (UCT). The second program, **IAWPRC**, is based on the model proposed by a task group of the International Association on Water Pollution Research and Control (IAWPRC), which in turn is based largely on the UCT model; the model and the program incorporate some further modifications by the UCT group.

The two programs are 'menu driven'. That is, for the most part the user selects a desired program option from a list of possible options displayed on the screen. This approach should be familiar to users of many commercial software packages and is adopted to simplify program operation, and to minimize the amount of typing by the user. A number of other features have been included in the programs in an effort to make these as 'user friendly' as possible; for example:

- The user will find that it is very difficult to upset program operation when inputting data. Protection against typing incorrect keys is built into the program wherever possible.
- A consequence of the complexity of the models is that they incorporate a large number of constants, kinetic, stoichiometric, and others. Default values for all the constants, wastewater characteristics, etc., are included. These values have been selected and calibrated for South African conditions; the default values can be updated by the user if necessary for the situation where the model is applied. In the sections of the manual dealing with kinetic and stoichiometric constants and wastewater characteristics, guidance is given as to situations in which the constants may require alteration.
- Installation of the programs is very simple. The size of memory in the computer determines the maximum number of reactors in a system that can be simulated. This is detected automatically by the program. Also, the program automatically detects the type of graphics adapter installed in the computer.

ABOUT THIS MANUAL

This manual has been designed to give complete support to users of the two simulation programs:

Chapter 2: Models and Wastewater Characteristics presents the models and briefly highlights the key features. The division of the influent wastewater COD and TKN fractions into various sub-fractions is described; e.g. biodegradable/unbiodegradable, soluble/particulate, and so on.

Chapter 3: Installing the Programs provides instructions on how to set up working programs from the 'distribution disk' accompanying the manual.

Chapter 4: Running the Programs demonstrates use of the programs in detail, and explains features of the various menus and sub-menus.

Chapter 5: Application and verification illustrates simulation of various experimental systems and includes comparison of the predicted and experimental data.

Chapter 6: Determination of model constants sets out procedures for determination of influent COD and TKN fractions and maximum specific growth rates for heterotrophs and autotrophs.

Appendix A: Retrieval of Diurnal Response Data shows how the data in a file of diurnal results may be re-arranged in a form suitable for off-line analysis. A program for making the conversion is provided; this allows the data to be imported into a spreadsheet package such as *Lotus 1-2-3* or Borland's *Quattro* should the user wish to, say, plot results in a specific format.

Appendix B: Model Representation in Matrix Format describes the matrix format used for presenting the models in Chapter 2.

Appendix C: Reactor Numbering Convention provides examples of the convention used for reactor numbering in the programs.

Appendix D: Symbols System sets out the basis for the system of symbols used in this manual.

CHAPTER 2

MODELS AND WASTEWATER CHARACTERISTICS

INTRODUCTION

Two activated sludge system models are included in the simulation package. These are **UCTOLD** and **IAWPRC**. In this Chapter the origins and key features of the two models are described briefly.

A matrix format is used for presenting the processes in the model, their kinetics and their stoichiometric interaction with the compounds. This format was recommended by the IAWPRC Task Group on Mathématical Modelling of Wastewater Treatment (Henze *et al.*, 1987; IAWPRC Task Group, 1987), and merits study; it provides an explicit 'fingerprint' of the processes and facilitates understanding of complex biological models. For those not familiar with this form of presentation, **Appendix B** describes elements of the matrix presentation method.

UNIVERSITY OF CAPE TOWN MODEL (UCTOLD)

Steady state aerobic model

The dynamic activated sludge system model incorporated in the **UCTOLD** program, evolved out of the steady state aerobic model of Marais and Ekama (1976). This steady state model constituted a development from a number of previous models for ***carbonaceous*** and ***nitrogenous*** material conversion and removal.

Carbonaceous material
Modelling of carbonaceous material conversion conformed principally to the proposals of McKinney (1962) and McKinney and Ooten (1969). These researchers proposed three mixed liquor volatile solids fractions; active, endogenous-inert and inert, the last named derived from the influent. Further, they proposed (1) a relationship between the mass of substrate utilized and the active mass of organisms synthesized, and an expression for the rate of organism mass synthesis; (2) an accumulation of volatile endogenous-inert solids (endogenous residue) associated with active mass loss due to endogenous respiration, and an expression for the rate of active mass loss; (3) a relationship between the oxygen demand and the organism active mass synthesized; (4) a relationship between the oxygen demand and the active mass loss due to endogenous respiration; and (5) an accumulation of inert volatile solids due to the presence of this material in the influent. Marais and Ekama (1976) accepted these proposals, with the exception of McKinney's proposal for the rate of synthesis of active mass. Instead they accepted Lawrence and McCarty's (1970) proposal linking the specific organism growth rate to the concentration of substrate surrounding the organism via the Monod relationship.

Marais and Ekama (1976) rejected the biochemical oxygen demand (BOD) as a suitable parameter for defining the carbonaceous material. Instead they accepted the electron donating capacity of the carbonaceous material in its equivalent form, the chemical oxygen demand (COD). They proposed that the influent COD be divided into three fractions; (1) biodegradable, (2) unbiodegradable particulate, and (3) unbiodegradable soluble. The growth and endogenous respiration processes were formulated in terms of the COD, or, where appropriate, in terms of an associated parameter the volatile suspended solids (VSS). Based on these considerations Marais and Ekama (1976) developed steady state equations for single and in-series completely mixed reactor activated sludge systems receiving a constant flow and load, i.e. equations for the active, endogenous and inert volatile solids concentrations and for oxygen consumption rates due to synthesis and endogenous respiration, these as functions of sludge age (organism retention time).

Nitrogenous material
Marais and Ekama (1976) proposed that the influent nitrogen (N) be divided into four fractions; (1) unbiodegradable soluble, (2) unbiodegradable particulate, (3) biodegradable organic, and (4) free and saline ammonia. Again they developed steady state equations for the utilization of ammonia for active cell mass synthesis, release of organic nitrogen due to endogenous respiration, conversion of biodegradable organic nitrogen to ammonia, incorporation of unbiodegradable nitrogen in inert material, and conversion of ammonia to nitrate (nitrification) as functions of sludge age. For the conversion of ammonia to nitrate they followed the Monod approach, as set out by Downing, Painter and Knowles (1964).

Summary
The most important features of the model of McKinney/Marais & Ekama can be summarized as follows:

In carbonaceous material conversion –

- subdivision of the influent COD into the three fractions;
- distinction between active, endogenous and inert sludge fractions;
- formulation of the biological reactions in terms of the active organism mass concentration; and
- linking oxygen utilization to the synthesis and endogenous processes.

In nitrogenous material conversion –

- subdivision of the influent nitrogen into the four fractions;
- formulation of the oxygen requirements for nitrification; and
- incorporation of nitrogen in the various sludge fractions.

Dynamic model

The steady state model, when it was applied to simulate single and in-series reactor systems under cyclic flow and load conditions, gave predictions that deviated significantly from that observed experimentally. When unaerated reactors were incorporated within the series for denitrification ('anoxic' reactors), even greater deviation from the model predictions were observed. Eventually these problems were resolved by the development of a dynamic model for the activated sludge system based on a mechanistic conceptualization of the aerobic and denitrification kinetic behaviour of the organisms in such a system. The development of this model is sketched below.

Progressively three models were produced, by Ekama, van Haandel and Marais (1979), Dold, Ekama and Marais (1980), and van Haandel, Ekama and Marais (1981); the last named model is the one incorporated in the UCTOLD program. Two key features of the model are of particular importance, namely, the ***bisubstrate*** and ***death-regeneration*** hypotheses.

Bisubstrate hypothesis

In nitrification/denitrification systems under constant flow and load, Stern and Marais (1974) had shown that in plugflow anoxic reactors upstream of the aerobic reactor (primary anoxic reactor) denitrification took place in two linear phases, a rapid first phase which persisted for a short period then terminated, and a second slow phase which continued for the retention time in the reactor. In plugflow anoxic reactors downstream of the aerobic reactor (secondary anoxic reactor), only one linear denitrification phase was operative, at a slow rate of about two thirds that of the slow second rate in the primary anoxic reactor. Ekama *et al.* (1979) hypothesized that the two linear phases in the primary anoxic reactor arose from the utilization of two biodegradable COD fractions in the influent, readily biodegradable (RBCOD) and slowly biodegradable (SBCOD); the first denitrification phase they connected to the RBCOD and the second to SBCOD. The slow rate single denitrification phase observed in secondary anoxic reactors they ascribed to endogenous mass loss. Based on these observations they developed a denitrification design procedure for in-series multi reactor systems under ***constant*** flow and load conditions. In 1980 Dold *et al.* incorporated RBCOD and SBCOD in an aerobic nitrification activated sludge kinetic model that gave very good predictions of system behaviour under cyclic flow and load conditions. Their study established the importance of RBCOD and SBCOD in describing the kinetic behaviour of the aerobic activated sludge system. In the model they postulated that the two COD substrate types, readily and slowly biodegradable, are acted on independently by the total active organism mass in the system:

> *Readily biodegradable substrate (RBCOD):* The RBCOD, they hypothesized, consists of small simple molecules that can pass directly through the cell wall (by passive or active uptake) for synthesis and oxidative metabolism by the organism. The rate of synthesis of active mass from the RBCOD is formulated according to the Monod equation linking the specific growth rate of the active mass to the RBCOD concentration in the liquid phase. The reaction rate is rapid.
>
> *Slowly biodegradable substrate (SBCOD):* The SBCOD, they hypothesized, consists of larger complex molecules that cannot pass directly through the cell wall. Utilization of this fraction involves four distinct phases; (1) enmeshment by the sludge mass, (2) adsorption, (3) extracellular enzymatic breakdown of the complex organic molecules to simpler components which pass directly through the cell wall, and (4) metabolism of the simpler components by the organism. For phase (1), the enmeshment of the SBCOD was assumed to be instantaneous and so was not explicitly modelled. For phases (3) and (4), the overall reaction rate is slow and appears to be limited by the rate of extracellular breakdown (hydrolysis) of the adsorbed SBCOD material [phase (3)] rather than the rate of metabolism of the hydrolysed material which passes through the cell wall [phase (4)]. Since hydrolysis is the rate limiting process, absorption and subsequent metabolism of the hydrolysis products were not explicitly modelled. Hence, the utilization of SBCOD could be modelled as a two-stage process; (1) adsorption of the material by the organism mass, and (2) enzymatic breakdown (hydrolysis) of the substrate. It is important to note that the hydrolysis products pass directly to the organism, not to the solution surrounding the organism.

Van Haandel *et al.* (1981) incorporated denitrification in the Dold *et al.* model to produce a general nitrification/denitrification activated sludge kinetic model. Van Haandel *et al.* found that the denitrification kinetic behaviour could be modelled in terms of RBCOD and SBCOD, and that the same formulations proposed by Dold *et al.* for RBCOD and SBCOD utilization under aerobic conditions could be used to model their utilization under anoxic conditions, except that the rate of SBCOD hydrolysis/utilization under anoxic conditions was reduced to about 1/3 that under aerobic conditions. Using the general nitrification/denitrification model, simulations of the denitrification response in primary and secondary anoxic plugflow reactors predicted near linear two phase denitrification behaviour in the primary anoxic reactor, and a single near linear phase in the secondary anoxic reactor, as observed by Stern and Marais (1974). Van Haandel *et al.* (1981) concluded that, in the plugflow primary anoxic reactor the first phase linear denitrification rate arose from utilization of RBCOD/adsorbed SBCOD and the second from utilization of adsorbed SBCOD. In the plugflow secondary anoxic reactor the single denitrification phase arose from utilization of adsorbed SBCOD generated from organism death.

Death-regeneration hypothesis

In earlier models the endogenous respiration concept was used to explain the phenomenon of active mass loss, i.e. a reduction in active volatile mass with time. This process was attributed to an energy requirement for organism maintenance, where a fraction of the organism mass disappears to provide energy for maintenance. However, in a reactor with sequential aerobic and anoxic states under constant flow and load conditions, in situations where the concentration of both oxygen and nitrate were zero for a period (i.e. anaerobic state), it was observed that when oxygen (or nitrate) was reintroduced the initial oxygen (or nitrate) demand increased significantly over that expected from the classical synthesis endogenous respiration approach. The increased oxygen (or nitrate) demand was found to be equal to the total demand that would have been expected if oxygen (or nitrate) had been available during the anaerobic period. The endogenous respiration approach could not explain this observation. Thus there was a need to develop an alternative model which could reflect the situation where aerobically (anoxically) generated organisms were placed in an anaerobic state for a short period. This led to the formulation of the death-regeneration model.

In the death-regeneration model an attempt is made to separate out the reactions which take place during the organism's 'death phase'. Disappearance of live active mass is hypothesized to be due to the net effect of death (natural or predation) and regeneration of organisms: On death the cell material is released through lysis; a fraction is unbiodegradable and remains as an unbiodegradable endogenous residue; the remaining fraction is biodegradable and becomes part of the SBCOD in the liquid, returning to the same cycle of adsorption, hydrolysis and, finally, synthesis of new cell mass (i.e. regeneration), giving rise to an associated oxygen demand. The classical endogenous oxygen demand thus in effect becomes a resynthesis oxygen demand. The main implication of this approach is that 'maintenance energy' *per se* (the oxygen requirement for maintenance) is considered to be so small that it can be lumped with, and completely swamped by, the oxygen demand for the synthesis of new cell mass from the lysed substrate.

Simulation studies utilizing the endogenous respiration and the death-regeneration approaches respectively showed that for dynamic modelling of aerobic and anoxic/aerobic activated sludge systems there is little difference in the predictions given by the two approaches. However, when modelling in-series reactor activated sludge systems that incorporate 'anaerobic' reactors (i.e. oxygen and nitrate limited conditions) the endogenous respiration approach gave very poor predictions, whereas the death-regeneration approach gave very good predictions. In the death-regeneration approach, when the nitrate concentration becomes zero in an unaerated reactor, synthesis ceases but organism death continues with the associated release of biodegradable substrate back into the liquid; this results in a build up of biodegradable material in the reactor. When the liquid passes to a downstream plugflow aerated reactor, the utilization of the accumulated biodegradable material results in an initial high oxygen utilization rate, much higher than that obtained when using the classical endogenous respiration approach.

Nitrogen transformations
Besides the carbonaceous conversion aspects described above, van Haandel *et al.* (1981) found that the bisubstrate death-regeneration approach could be integrated in a consistent manner with the transformations of nitrogen, in synthesis, death-regeneration, nitrification and denitrification.

UCTOLD process model description

The process model describes the processes, their kinetics and the compounds on which they act and defines the behaviour at a single point in a system. The process model incorporated in the **UCTOLD** program is presented in matrix form in Table 2.1. This matrix provides a quantitative description of interrelationships between the processes (j) and associated compounds (i). Details of the symbol system used in this manual are given in **Appendix D**. The stoichiometric, kinetic and switching function constants in the **UCTOLD** process model are defined in the section **'List of Symbols'** and their numerical values are listed in Table 2.2; these are the default values for use in the program. In **Chapter 6**, guidance is given on the determination of some of the constants. The switching functions used in the matrix are formulated in Table 2.3. A short description of the processes and their stoichiometric interactions with the compounds, is set out below.

Processes 1 and 2 ***Aerobic growth of heterotrophs on readily biodegradable COD (RBCOD):*** These two processes are responsible for removal of RBCOD under aerobic conditions. A fraction of the RBCOD is used for the production (growth) of heterotrophic biological (active) organism mass (synthesis) and the balance is oxidized for energy giving rise to an associated synthesis oxygen demand. The growth is modelled using Monod kinetics. In Process 1, concomitant with growth, ammonia is used as a nitrogen source for synthesis, and there is an associated alkalinity change. Should ammonia become limiting, nitrate can serve as an alternative nitrogen source for synthesis, Process 2. This is accomplished by incorporating switching functions dependent on the ammonia concentration, which switch Process 1 'off' and Process 2 'on' when ammonia reduces to near zero but nitrate is available as a nitrogen source. Should both ammonia and nitrate reduce to zero, growth Processes 1 and 2 are switched off. Additional switching functions in both Processes 1 and 2 ensure that these processes only operate if oxygen is present as electron acceptor - as the oxygen concentration reduces to zero there is a switch from aerobic to anoxic growth (Processes 3 and 4) with nitrate as electron acceptor.

Processes 3 and 4 ***Anoxic growth of heterotrophs on readily biodegradable COD (RBCOD):*** In the absence of oxygen, the heterotrophic organism population is capable of using nitrate, if available, as electron acceptor with RBCOD as substrate. In these two processes, a fraction of the RBCOD is used for production (growth) of heterotrophic biological (active) organism mass and the balance is oxidized giving rise to reduction of nitrate to nitrogen gas. The anoxic growth is modelled using the same Monod kinetics used for aerobic growth (Processes 1 and 2). As in aerobic growth (with RBCOD), ammonia (Process 3) or nitrate (Process 4) can serve as nitrogen source for cell synthesis; this is accommodated through switching

Table 2.1: Matrix representation of UCT model

j ↓ PROCESS / COMPOUND i →	1 Z_{BH}	2 Z_{BA}	3 Z_E	4 Z_I	5 S_{ads}	6 S_{enm}	7 N_{obp}	8 S_{bs}	9 N_a	10 N_{obs}	11 N_{o3}	12 Alk	13 S_{us}	14 O	PROCESS RATES, ρ_j
1 Aerobic growth of Z_{BH} on S_{bs} with N_a	1							$-1/Y_{ZH}$	$-f_{ZB,N}$			$-f_{ZB,N}/14$		$-\frac{1-Y_{ZH}}{Y_{ZH}}$	$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}H\ Air\\ on\end{matrix}\right]\left[\begin{matrix}N_a\\ Limit\end{matrix}\right] Z_{BH}$
2 Aerobic growth of Z_{BH} on Sbs with N_{o3}	1							$-1/Y_{ZH}$			$-f_{ZB,N}$	$f_{ZB,N}/14$		$-\frac{1-Y_{ZH}}{Y_{ZH}}$	$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}H\ Air\\ On\end{matrix}\right]\left[\begin{matrix}1-N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}$
3 Anoxic growth of Z_{BH} on S_{bs} with N_a	1							$-1/Y_{ZH}$	$-f_{ZB,N}$		$-\frac{1-Y_{ZH}}{2.86Y_{ZH}}$	A^*			$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}H\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}$
4 Anoxic growth of Z_{BH} on S_{bs} with N_{o3}	1							$-1/Y_{ZH}$			B^*	C^*			$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}H\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}1-N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}$
5 Aerobic growth of Z_{BH} on S_{ads} with N_a	1				$-1/Y_{ZH}$				$-f_{ZB,N}$			$-f_{ZB,N}/14$		$-\frac{1-Y_{ZH}}{Y_{ZH}}$	$K_{MP}\left[\frac{(S_{ads}/Z_{BH})}{K_{SP}+(S_{ads}/Z_{BH})}\right]\left[\begin{matrix}H\ Air\\ On\end{matrix}\right]\left[\begin{matrix}N_a\\ Limit\end{matrix}\right] Z_{BH}$
6 Aerobic growth of Z_{BH} on S_{ads} with N_{o3}	1				$-1/Y_{ZH}$						$-f_{ZB,N}$	$f_{ZB,N}/14$		$-\frac{1-Y_{ZH}}{Y_{ZH}}$	$K_{MP}\left[\frac{(S_{ads}/Z_{BH})}{K_{SP}+(S_{ads}/Z_{BH})}\right]\left[\begin{matrix}H\ Air\\ On\end{matrix}\right]\left[\begin{matrix}1-N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}$
7 Anoxic growth of Z_{BH} on S_{ads} with N_a	1				$-1/Y_{ZH}$				$-f_{ZB,N}$		$-\frac{1-Y_{ZH}}{2.86Y_{ZH}}$	A^*			$K_{MP}\left[\frac{(S_{ads}/Z_{BH})}{K_{SP}+(S_{ads}/Z_{BH})}\right]\left[\begin{matrix}H\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}\,\eta_G$
8 Anoxic growth of Z_{BH} on S_{ads} with N_{o3}	1				$-1/Y_{ZH}$						B^*	C^*			$K_{MP}\left[\frac{(S_{ads}/Z_{BH})}{K_{SP}+(S_{ads}/Z_{BH})}\right]\left[\begin{matrix}H\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}1-N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}\,\eta_G$
9 Death of Z_{BH}	-1		f_E			$1-f_E$	$f_{ZB,N}$ $-f_E f_{ZE,N}$								$b_H Z_{BH}$
10 Adsorption of S_{enm}					1	-1									$K_A S_{enm} Z_{BH}(f_{MA} - S_{ads}/Z_{BH})$
11 Hydrolysis of N_{obp}							-1			1					$(\rho_5+\rho_6+\rho_7+\rho_8)(N_{obp}/S_{ads})$
12 Ammonification of N_{obs}									1	-1		1/14			$K_R N_{obs} Z_{BH}$
13 Aerobic growth of Z_{BA}		1							$-1/Y_{ZA}$ $-f_{ZB,N}$		$1/Y_{ZA}$	$-f_{ZB,N}/14$ $-1/(7Y_{ZA})$		$-\frac{4.57-Y_{ZA}}{Y_{ZA}}$	$\hat{\mu}_A\left[\frac{N_a}{K_{SA}+N_a}\right]\left[\begin{matrix}A\ Air\\ On\end{matrix}\right] Z_{BA}$
14 Death of Z_{BA}		-1	f_E			$1-f_E$	$f_{ZB,N}$ $-f_E f_{ZE,N}$								$b_A Z_{BA}$
	Biological (active) heterotrophic mass - M (COD) L^{-3}	Biological (active) autotrophic mass - M (COD) L^{-3}	Endogenous mass - M (COD) L^{-3}	Inert mass - M (COD) L^{-3}	Adsorbed slowly biodegradable substrate - M (COD) L^{-3}	Enmeshed slowly biodegradable substrate - M (COD) L^{-3}	Biodegradable particulate organic nitrogen - M (N) L^{-3}	Readily biodegradable (soluble) substrate - M (COD) L^{-3}	Ammonia nitrogen - M (N) L^{-3}	Biodegradable soluble organic nitrogen - M (N) L^{-3}	Nitrate nitrogen - M (N) L^{-3}	Alkalinity - molar units	Unbiodegradable soluble substrate - M (COD) L^{-3}	Oxygen - M (-COD) L^{-3}	

KEY

$$A^* = \frac{1-Y_{ZH}}{14+2.86\,Y_{ZH}} - f_{ZB,N}/14$$

$$B^* = \frac{1-Y_{ZH}}{2.86\,Y_{ZH}} - f_{ZB,N}$$

$$C^* = \frac{1-Y_{ZH}}{14+2.86\,Y_{ZH}} + f_{ZB,N}/14$$

Table 2.2: Default values for kinetic, stoichiometric and switching function constants in the UCT model

SYMBOL	VALUE	UNITS
Kinetic parameters (20°C)		
$\hat{\mu}_H$	1,5 - 3,5	d^{-1}
K_{SH}	5,0	g COD m^{-3}
K_{MP}	1,35	g COD (g cell COD.d)$^{-1}$
K_{SP}	0,027	g COD (g cell COD)$^{-1}$
b_H	0,62	d^{-1}
K_A	0,17	g COD (g cell COD.d)$^{-1}$
K_R	0,032	m^3 (g cell COD.d)$^{-1}$
$\hat{\mu}_A$	0,2 - 0,75	d^{-1}
K_{SA}	1,0	g NH_3-N m^{-3}
b_A	0,04	d^{-1}
η_G	0,33	-
Stoichiometric parameters		
Y_{ZH}	0,666	g cell COD (g COD utilized)$^{-1}$
Y_{ZA}	0,15	g cell COD (g N utilized)$^{-1}$
f_{MA}	1,00	g COD (g cell COD)$^{-1}$
f_E	0,08	-
$f_{ZB,N}$	0,068	g N (g COD)$^{-1}$ in active mass
$f_{ZE,N}$	0,068	g N (g COD)$^{-1}$ in endogenous mass
Switching function parameters		
K_{OH}	0,002	g O_2 m^{-3}
K_{OA}	0,002	g O_2 m^{-3}
K_{HA}	0,01	g NH_3-N m^{-3}
K_{NO}	0,1	g NO_3-N m^{-3}
Arrhenius temperature dependency constants		
$K_{i,T} = K_{i,20}\,\theta^{(T-20)}$		
		where $K_{i,T} = K_i$ at T°C θ = Arrhenius constant
$\hat{\mu}_H$	1,200	-
K_{SH}	1,000	-
K_{MP}	1,080	-
K_{SP}	0,910	-
b_H	1,029	-
K_A	1,029	-
f_{MA}	1,000	-
K_R	1,029	-
$\hat{\mu}_A$	1,123	-
K_{SA}	1,123	-
b_A	1,029	-

Table 2.3: Switching functions used in UCT model (Table 2.1) and IAWPRC Task Group model (Table 2.4)

SWITCHING FUNCTION	FORMULATION
$\begin{bmatrix} \text{H Air} \\ \text{On} \end{bmatrix}$	$\dfrac{O}{K_{OH} + O}$
$\begin{bmatrix} \text{H Air} \\ \text{Off} \end{bmatrix}$	$\dfrac{K_{OH}}{K_{OH} + O}$
$\begin{bmatrix} \text{A Air} \\ \text{On} \end{bmatrix}$	$\dfrac{O}{K_{OA} + O}$
$\begin{bmatrix} N_a \\ \text{Limit} \end{bmatrix}$	$\dfrac{N_a}{K_{HA} + N_a}$
$\begin{bmatrix} 1\text{-}N_a \\ \text{Limit} \end{bmatrix}$	$\dfrac{K_{HA}}{K_{HA} + N_a}$
$\begin{bmatrix} N_{o3} \\ \text{Limit} \end{bmatrix}$	$\dfrac{N_{o3}}{K_{NO} + N_{o3}}$

functions. Further switching functions in both Processes 3 and 4 ensure that the anoxic growth rates decrease to zero at low nitrate concentrations.

Processes 5 and 6 *Aerobic growth of heterotrophs on adsorbed slowly biodegradable COD (SBCOD):* In these two processes the SBCOD, which has been adsorbed on the organism (Process 10), is utilized under aerobic conditions. This utilization consists of two steps, hydrolysis of the adsorbed SBCOD and direct utilization of the hydrolysis products for the production (growth) of active organism mass (synthesis) and its associated oxygen demand. Since the rate limiting step is hydrolysis, only this step is modelled, using Levenspiel's surface reaction kinetics (Dold *et al.*, 1980). As in aerobic growth on RBCOD, ammonia (Process 5) or nitrate (Process 6) can serve as a nitrogen source for cell synthesis; this is accomplished by incorporating switching functions dependent on the ammonia concentration. Also, switching functions ensure that both processes operate only if oxygen is present as electron acceptor; at low oxygen concentrations there is a switch to anoxic growth (Processes 7 and 8) with nitrate as electron acceptor.

Processes 7 and 8 *Anoxic growth of heterotrophs on adsorbed slowly biodegradable COD (SBCOD):* These processes are modelled in the same way as for the aerobic growth Processes 5 and 6 except that in the absence of oxygen, nitrate serves as an alternative electron acceptor and that the Levenspiel surface reaction for the hydrolysis/utilization kinetic rate expression is multiplied by the factor η_G. Again, either ammonia (Process 7) or nitrate (Process 8) can serve as nitrogen source for cell synthesis. Switching functions in both Processes 7 and 8 ensure that the anoxic growth rates decrease to zero at low nitrate concentrations.

Process 9 *Death of heterotrophs:* The process is modelled according to the death-regeneration hypothesis. That is, the heterotrophic organism mass dies at a certain rate per unit organism mass; a portion of the material from the death is unbiodegradable particulate and adds to the unbiodegradable endogenous residue while the remainder adds to the pool of SBCOD. Nitrogen associated with the biodegradable portion adds to the pool of particulate organic nitrogen. The process of organism death is assumed to continue under aerobic, anoxic and anaerobic conditions.

Process 10 ***Adsorption of slowly biodegradable COD (SBCOD):*** SBCOD is assumed to be enmeshed in the sludge mass immediately on contact with mixed liquor. The adsorption process transfers the enmeshed SBCOD to the adsorbed SBCOD.

Process 11 ***Hydrolysis of particulate organic nitrogen:*** Biodegradable particulate organic nitrogen is broken down to soluble organic nitrogen at a rate linked directly to the rate of hydrolysis/utilization of adsorbed SBCOD under both aerobic (Processes 5 and 6) and anoxic (Processes 7 and 8) conditions. The product of breakdown adds to the pool of soluble organic nitrogen.

Process 12 ***Ammonification of soluble organic nitrogen:*** Biodegradable soluble organic nitrogen is converted to free and saline ammonia, a process mediated by the heterotrophic biological active mass. Hydrogen ions consumed in the conversion process results in an alkalinity change.

Process 13 ***Aerobic growth of autotrophs:*** In this process ammonia is oxidized to nitrate via a single step resulting in production (growth) of autotrophic biological active mass and giving rise to an associated nitrification oxygen demand. The process requires the presence of oxygen; a switching function ensures that the process operates only if oxygen is present. This process has a marked effect on the predicted total oxygen demand; the effect on total MLVSS is small as the yield of autotrophic nitrifiers is low. As with the growth of heterotrophs, ammonia nitrogen is incorporated in the new cells and the alkalinity is affected.

Process 14 ***Death of autotrophs:*** The process parallels that for heterotrophs (Process 9).

UCTOLD system model

The process model of the compounds, processes and rates presented above defines the behaviour at a single point in a system. To obtain the response of the system the following need to be incorporated: System configuration (single or multiple reactors), system type (continuously fed, batch, etc.), hydraulic mixing regime in each reactor (plug flow or completely mixed), and recycle flows between reactors. These, together with the process model, form the system model. (Details on how to develop the system model from the process model are given in **Appendix B**). The system model in the **UCTOLD** program presented with this manual is for the resolution of the response of a system consisting of single or in-series completely mixed reactors; with up to two inter-reactor recycles and a single underflow recycle, all opposite to the direction of flow; any reactor can be aerated or unaerated; influent flow and COD and nitrogen loads can be constant or cyclic; sludge draw off to control the sludge age takes place from the reactor immediately upstream of the settling tank; solids separation in the settling tank is assumed to be 100 percent; and the mass of sludge in the settler is assumed negligible.

UCTOLD system model solution

Solution techniques and algorithms used in the **UCTOLD** computer program, to solve the system model within the boundaries described above, are set out in Billing (1987) and Billing and Dold (1988a, b and c).

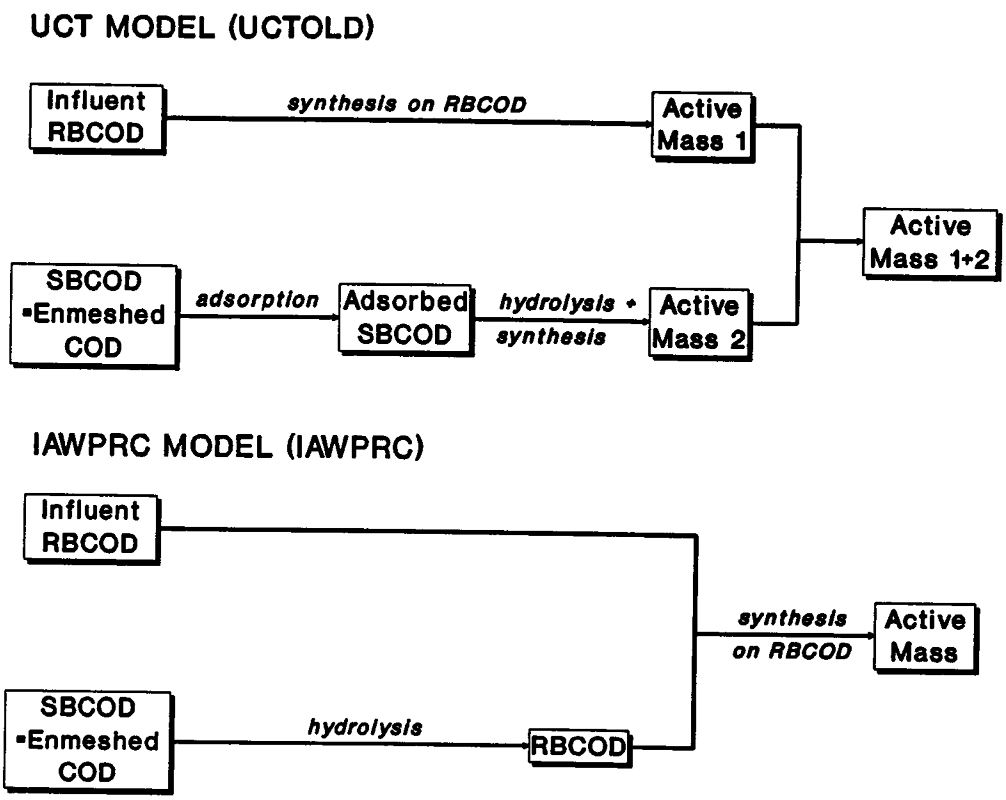
INTERNATIONAL ASSOCIATION ON WATER POLLUTION RESEARCH AND CONTROL MODEL (IAWPRC)

IAWPRC model origins

In 1982 the IAWPRC appointed a Task Group to review modelling of activated sludge systems. Their deliberations culminated in a proposal for an activated sludge model (Henze *et al.*, 1987; IAWPRC, 1987). This model is based largely on the existing model of the UCT Group, but with two principal changes, these are, in the enmeshment-adsorption (storage) and in the solubilization (hydrolysis) hypotheses.

The Task Group agreed with the proposal in the UCT model that the particulate material is enmeshed virtually instantaneously by the sludge mass. However, they rejected the UCT proposal that the enmeshed slowly biodegradable COD (SBCOD) is adsorbed on the organism mass. Instead they proposed that the enmeshed SBCOD is hydrolysed to readily biodegradable COD (RBCOD) by the action of extracellular enzymes secreted by the organism mass to the bulk liquid. The SBCOD thus hydrolysed to RBCOD adds to the RBCOD derived from the influent and the total is then available for synthesis. With these modifications the need to model the rate of adsorption (as in the UCT model) falls away.

The differences between the two approaches with respect to substrate utilization are set out in Fig 2.1. When appropriately calibrated the two approaches give virtually identical predictions, see **Chapter 5**.



<u>Fig 2.1</u>: Diagram illustrating similarities/differences between the UCT and IAWPRC approaches with respect to hydrolysis/synthesis of carbonaceous material.

A second aspect in which the Task Group model differs from the UCT model is that in the UCT model provision is made for nitrate to serve as a nitrogen source for cell synthesis if ammonia is not available. This provision was not included in the original IAWPRC Task Group model (Dold and Marais, 1985), but is included in the version incorporated in this manual and in the computer program called **IAWPRC**.

IAWPRC process model description

The process model is presented in matrix format in Table 2.4. This provides a quantitative description of the model and shows the inter-relationships between the processes (j) and associated compounds (i). Details of the symbol system used in this manual are given in **Appendix D**. The switching functions used in the matrix are formulated in Table 2.3. The stoichiometric, kinetic and switching function constants in the IAWPRC process model are defined in the section **'List of Symbols'** and the numerical values are listed in Table 2.5; these are the default values for use in the program. In **Chapter 6**, guidance is given on the determination of some of the constants. A short description of the processes and the stoichiometric interactions with the compounds is set out below.

In the **IAWPRC** model, heterotrophic growth takes place only on readily biodegradable COD (RBCOD), Processes 1, 2, 3 and 4.

Processes 1 and 2 ***Aerobic growth of heterotrophs:*** Processes 1 and 2 are responsible for removal of RBCOD [from the influent and generated from slowly biodegradable COD (SBCOD) hydrolysis, Process 6] under aerobic conditions. A fraction of the RBCOD is used for the production (growth) of heterotrophic biological active organism mass (synthesis) and the balance is oxidized for energy giving rise to an associated oxygen demand. The growth is modelled using Monod kinetics. In Process 1, concomitant with growth, ammonia is used as a nitrogen source for synthesis, and there is an associated alkalinity change. Should ammonium become limiting, nitrate can serve as an alternative nitrogen source for synthesis, Process 2, and there is an associated alkalinity change. The switch from ammonia to nitrate as nitrogen source for synthesis is accomplished by incorporating switching functions dependent on the ammonia concentration, which switch Process 1 'off' and Process 2 'on' when ammonia reduces to near zero but nitrate is available. Should both ammonia and nitrate reduce to zero, growth Processes 1 and 2 are switched off. Further switching functions in both Processes 1 and 2 ensure that these processes only operate if oxygen is present as electron acceptor; at low oxygen concentrations there is a switch from aerobic to anoxic growth (Processes 3 and 4) with nitrate as electron acceptor.

Processes 3 and 4 ***Anoxic growth of heterotrophs:*** In the absence of oxygen the heterotrophic organism population is capable of using nitrate, if available, as electron acceptor with RBCOD as substrate. In these two processes, a fraction of the RBCOD is used for production (growth) of active organism mass and the balance gives rise to reduction of nitrate to nitrogen gas with an associated alkalinity change. The anoxic growth is modelled using the same Monod kinetics used for aerobic growth (Processes 1 and 2) except that the kinetic rate expression is multiplied by the factor η_G. (For the present it is assumed that the potential growth rate is the same under anoxic as under aerobic conditions, i.e. $\eta_G = 1$. The limitation in growth rate under anoxic conditions can arise in hydrolysis, see Process 6). As for aerobic growth with RBCOD, ammonia (Process 3) or nitrate (Process 4) can serve as nitrogen source for cell synthesis; this is accommodated through switching functions. Further switching functions in both Processes 3 and 4 ensure that the anoxic growth rates decrease to zero at low nitrate concentrations.

Table 2.4: Matrix representation of IAWPRC Task Group model

j / Compound i → Process	1 Z_{BH}	2 Z_{BA}	3 Z_E	4 Z_I	5 S_{enm}	6 N_{obp}	7 S_{bs}	8 N_a	9 N_{obs}	10 N_{o3}	11 Alk	12 S_{us}	13 O	PROCESS RATES, ρ_j
1 Aerobic growth of Z_{BH} with N_a	1						$-1/Y_{ZH}$	$-f_{ZB,N}$			$-f_{ZB,N}/14$		$-\frac{1-Y_{ZH}}{Y_{ZH}}$	$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}N\ Air\\ on\end{matrix}\right]\left[\begin{matrix}N_a\\ Limit\end{matrix}\right] Z_{BH}$
2 Aerobic growth of Z_{BH} with N_{o3}	1						$-1/Y_{ZH}$			$-f_{ZB,N}$	$f_{ZB,N}/14$		$-\frac{1-Y_{ZH}}{Y_{ZH}}$	$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}N\ Air\\ On\end{matrix}\right]\left[\begin{matrix}1-N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}$
3 Anoxic growth of Z_{BH} with N_a	1						$-1/Y_{ZH}$	$-f_{ZB,N}$		$-\frac{1-Y_{ZH}}{2.86Y_{ZH}}$	A^*			$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}N\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}\,\eta_G$
4 Anoxic growth of Z_{BH} with N_{o3}	1						$-1/Y_{ZH}$			B^*	C^*			$\hat{\mu}_H \left[\frac{S_{bs}}{K_{SH}+S_{bs}}\right]\left[\begin{matrix}N\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}1-N_a\\ Limit\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right] Z_{BH}\,\eta_G$
5 Death of Z_{BH}	-1		f_E		$1-f_E$	$f_{ZB,N}$ $-f_E f_{ZE,N}$								$b_H Z_{BH}$
6 Hydrolysis of S_{enm}					-1		1							$K_H \left[\frac{(S_{enm}/Z_{BH})}{K_X+(S_{enm}/Z_{BH})}\right]\left(\left[\begin{matrix}N\ Air\\ On\end{matrix}\right]+\eta_S\left[\begin{matrix}N\ Air\\ Off\end{matrix}\right]\left[\begin{matrix}N_{o3}\\ Limit\end{matrix}\right]\right) Z_{BH}$
7 Hydrolysis of N_{obp}						-1			1					$\rho_6 (N_{obp}/S_{enm})$
8 Ammonification of N_{obs}								1	-1		1/14			$K_R N_{obs} Z_{BH}$
9 Aerobic growth of Z_{BA}		1						$-1/Y_{ZA}$ $-f_{ZB,N}$		$1/Y_{ZA}$	$-f_{ZB,N}/14$ $-1/(7Y_{ZA})$		$-\frac{4.57-Y_{ZA}}{Y_{ZA}}$	$\hat{\mu}_A \left[\frac{N_a}{K_{SA}+N_a}\right]\left[\begin{matrix}A\ Air\\ On\end{matrix}\right] Z_{BA}$
10 Death of Z_{BA}		-1	f_E		$1-f_E$	$f_{ZB,N}$ $-f_E f_{ZE,N}$								$b_A Z_{BA}$
	Biological (active) heterotrophic mass - M (COD) L^{-3}	Biological (active) autotrophic mass - M (COD) L^{-3}	Endogenous mass - M (COD) L^{-3}	Inert mass - M (COD) L^{-3}	Enmeshed slowly biodegradable substrate - M (COD) L^{-3}	Biodegradable particulate organic nitrogen - M (N) L^{-3}	Readily biodegradable (soluble) substrate - M (COD) L^{-3}	Ammonia nitrogen - M (N) L^{-3}	Biodegradable soluble organic nitrogen - M (N) L^{-3}	Nitrate nitrogen - M (N) L^{-3}	Alkalinity - molar units	Unbiodegradable soluble substrate - M (COD) L^{-3}	Oxygen - M (-COD) L^{-3}	

KEY

$$A^* = \frac{1-Y_{ZH}}{14 \cdot 2.86\, Y_{ZH}} - f_{ZB,N}/14$$

$$B^* = \frac{1-Y_{ZH}}{2.86\, Y_{ZH}} - f_{ZB,N}$$

$$C^* = \frac{1-Y_{ZH}}{14 \cdot 2.86\, Y_{ZH}} + f_{ZB,N}/14$$

Table 2.5: Default values for kinetic, stoichiometric and switching function constants in the IAWPRC Task Group model

SYMBOL	VALUE	UNITS
Kinetic parameters (20°C)		
$\hat{\mu}_H$	2,4 - 5,0	d^{-1}
K_{SH}	5,0	g COD m^{-3}
K_H	2,03	g COD (g cell COD.d)$^{-1}$
K_X	0,027	g COD (g cell COD)$^{-1}$
b_H	0,62	d^{-1}
K_R	0,032	m^3 (g cell COD.d)$^{-1}$
$\hat{\mu}_A$	0,2 - 0,75	d^{-1}
K_{SA}	1,0	g NH_3-N m^{-3}
b_A	0,04	d^{-1}
η_S	0,33	-
η_G	1,00	-
Stoichiometric parameters		
Y_{ZH}	0,666	g cell COD (g COD utilized)$^{-1}$
Y_{ZA}	0,15	g cell COD (g N utilized)$^{-1}$
f_E	0,08	-
$f_{ZB,N}$	0,068	g N (g COD)$^{-1}$ in active mass
$f_{ZE,N}$	0,068	g N (g COD)$^{-1}$ in endogenous mass
Switching function parameters		
K_{OH}	0,002	g O_2 m^{-3}
K_{OA}	0,002	g O_2 m^{-3}
K_{HA}	0,01	g NH_3-N m^{-3}
K_{NO}	0,1	g NO_3-N m^{-3}
Arrhenius temperature dependency constants		
$K_{i,T} = K_{i,20}\,\theta^{(T-20)}$		
	where $K_{i,T}$ = K_i at T°C; θ = Arrhenius constant	
$\hat{\mu}_H$	1,200	-
K_{SH}	1,000	-
K_H	1,080	-
K_X	0,910	-
b_H	1,029	-
K_R	1,029	-
$\hat{\mu}_A$	1,123	-
K_{SA}	1,123	-
b_A	1,029	-

Process 5 *Death of heterotrophs:* The process is modelled according to the death-regeneration hypothesis. That is, the heterotrophic organism mass dies at a certain rate; a portion of the material from death is unbiodegradable and adds to the unbiodegradable endogenous residue while the remainder adds to the pool of SBCOD. Nitrogen associated with the SBCOD becomes available as particulate organic nitrogen. The process of organism death is assumed to continue under aerobic, anoxic and anaerobic conditions.

Process 6 *'Hydrolysis' of slowly biodegradable COD (SBCOD):* SBCOD enmeshed in the sludge mass is broken down extracellularly, with the products of breakdown adding to the pool of RBCOD available to the organisms for growth (Processes 1, 2, 3 and 4). This 'hydrolysis' process is modelled on the basis of Levenspiel's surface reaction kinetics, and occurs only under aerobic or anoxic conditions; the rate of hydrolysis under anoxic conditions is reduced compared to the rate under aerobic conditions by multiplying the aerobic hydrolysis rate by the factor η_s under anoxic conditions. Hydrolysis does not occur under anaerobic conditions; a switching functions ensure that the hydrolysis process only operates if oxygen or nitrate is present.

Process 7 *'Hydrolysis' of particulate organic nitrogen:* Biodegradable particulate organic nitrogen is broken down to soluble organic nitrogen at a rate defined by the hydrolysis reaction for SBCOD (Process 6 above). The product of breakdown adds to the pool of soluble organic nitrogen.

Process 8 *Ammonification of soluble organic nitrogen:* Biodegradable soluble organic nitrogen is converted to free and saline ammonia, a process mediated by the active heterotrophs. Hydrogen ions consumed in the conversion process results in an alkalinity change.

Process 9 *Aerobic growth of autotrophs:* In this process ammonia is oxidized to nitrate via a single step resulting in production (growth) of autotroph active mass and giving rise to an associated oxygen demand. The process requires the presence of oxygen; a switching function ensures that the process only operates if oxygen is present. This process has a marked effect on the alkalinity and the total oxygen demand; the effect on total MLVSS is small as the yield of autotrophic nitrifiers is low. As with the growth of heterotrophs, ammonia nitrogen is incorporated in the new cells and the alkalinity is affected.

Process 10 *Death of autotrophs:* The process parallels that for heterotrophs (Process 5).

IAWPRC system model

The process model of the compounds presented above defines the behaviour at a single point in a system. To obtain the response of the system a number of factors need to be incorporated. The reader is referred to the previous section **'UCTOLD system model'** where consideration is given to these factors, and conditions for application of the UCT model are set out; these apply also to the IAWPRC model.

IAWPRC system model solution

The same solution techniques and algorithms employed in the UCTOLD program are used in the IAWPRC program; details of these are set out in Billing (1987) and Billing and Dold (1988a, b and c).

INFLUENT WASTEWATER CHARACTERISTICS

Carbonaceous material

Characterization of the carbonaceous material in the influent is in terms of the chemical oxygen demand (COD). For use in the models it is necessary to specify the magnitudes of the various fractions of the influent COD. (Details of the symbol system used in this manual are given in **Appendix D**). Research by the UCT Group has indicated a division shown diagrammatically in Fig 2.2.

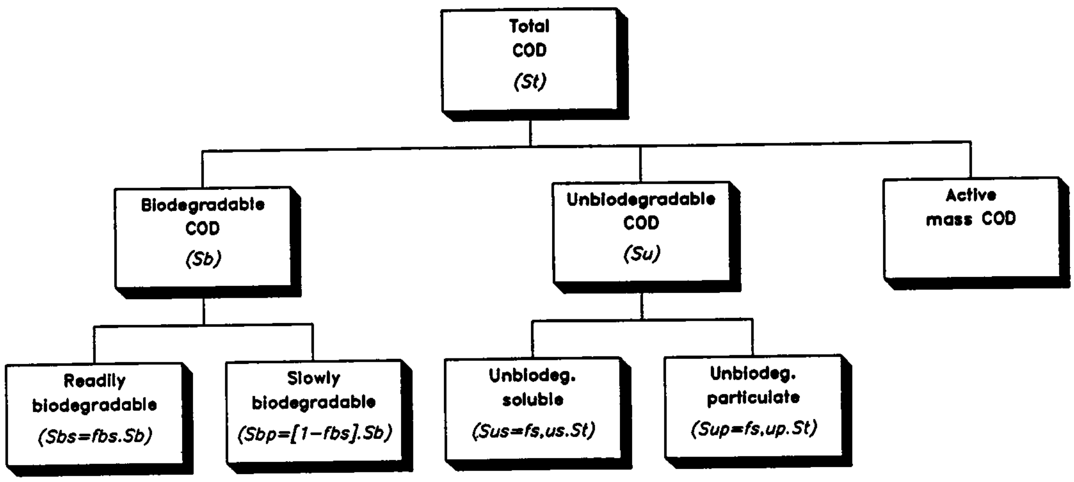


Fig 2.2: Division of the influent COD into its constituent fractions.

Biodegradable and unbiodegradable fractions
The first division of the influent COD (S_{ti}) is into biodegradable COD (S_{bi}) and unbiodegradable COD (S_{ui}). Each of these fractions is subdivided further into two sub-fractions:

Unbiodegradable sub-fractions: The influent unbiodegradable COD (S_{ui}) is divided into two fractions, unbiodegradable **soluble** COD (S_{usi}) and unbiodegradable **particulate** COD (S_{upi}). Both are hypothesized to be unaffected by biological action in the system. The S_{us} passes out in the secondary settling tank overflow; the S_{up} is enmeshed in the sludge mass and accumulates in the system as inert VSS (Z_I in COD units). At steady state the mass of S_{up} entering the system in the influent will be balanced by the mass of Z_I leaving via the sludge wastage stream. From a mass balance, the mass of Z_I in the system will equal the influent mass of S_{up} per day multiplied by the system sludge age.

Biodegradable sub-fractions: The influent biodegradable COD (S_{bi}) is divided into two fractions, readily biodegradable COD (RBCOD, S_{bsi}) and slowly biodegradable COD (SBCOD, S_{bpi}). The RBCOD is hypothesized to consist of simple soluble molecules that can be absorbed readily by the organism and metabolized for energy and synthesis, whereas the SBCOD is assumed to be made up of particulate/colloidal/complex organic molecules that require extracellular enzymatic breakdown prior to absorption and utilization.

Quantifying the division
From practical considerations, the division of the influent wastewater COD into fractions is conventionally defined by the fractional constants, $f_{S,us}$, $f_{S,up}$ and f_{bs}:

$f_{S,us}$ = fraction of the **total** influent COD which is unbiodegradable soluble;

$f_{S,up}$ = fraction of the **total** influent COD which is unbiodegradable particulate;

f_{bs} = fraction of the **biodegradable** influent COD which is readily biodegradable.

Using these fractional constants the division of the influent COD into the various sub-fractions may be expressed as follows:

$$S_{usi} = f_{S,us} \cdot S_{ti} \tag{2.1}$$

$$S_{upi} = f_{S,up} \cdot S_{ti} \tag{2.2}$$

The biodegradable influent COD (S_{bi}) is given by the difference of the total influent COD (S_{ti}) and the sum of S_{usi} and S_{upi}:

$$\begin{aligned} S_{bi} &= S_{ti} - (S_{usi} + S_{upi}) \\ &= S_{ti}\,(1 - f_{S,us} - f_{S,up}) \\ &= S_{bsi} + S_{bpi} \end{aligned} \tag{2.3}$$

where

$$S_{bsi} = f_{bs} \cdot S_{bi} \tag{2.4}$$

$$S_{bpi} = (1 - f_{bs} \cdot S_{bi}) \tag{2.5}$$

To specify the wastewater fully, values must be assigned to the total influent COD concentration (S_{ti}) and the fractional constants, $f_{S,us}$, $f_{S,up}$ and f_{bs}. The magnitudes of these constants will vary from influent to influent, and differ for a raw and a primary settled wastewater. In the program, default values are provided for raw and settled wastewaters; these are approximate average values observed for municipal wastewaters in South Africa. Experimental procedures for estimating these fractional constants are set out by Ekama, Dold and Marais (1986) and are reviewed in **Chapter 6**.

The IAWPRC Task Group proposed a further influent COD fraction, the COD of active aerobic organisms present in the influent. This COD fraction is shown in Fig 2.2 and incorporated in both models. In the models, quantification of this COD fraction (S_{ZBi}) is via the fractional constant $f_{S,ZB}$ where,

$$S_{ZBi} = f_{S,ZB} \cdot S_{ti}$$

For municipal wastewaters in South Africa it appears that the fractional constant $f_{S,ZB}$ can be taken as zero as the sewers generally are anaerobic. However, in cold temperate climates and with sewer systems that are aerated the influent active organism COD fraction may be significant. At present its magnitude can be assessed only by simulation, comparing the predicted response with the experimental response on laboratory-scale units treating the influent (Sollfrank and Gujer, 1991). Should the influent active organism COD fraction be significant, it is subtracted from the influent total COD and adds to the active organism mass in the reactor.

The remaining influent COD is divided into the biodegradable/unbiodegradable fractions as discussed above.

Nitrogenous material

Characterization of the nitrogenous material in the influent is in terms of the total Kjeldahl nitrogen (TKN). It is necessary to specify the magnitudes of the various fractions of the influent TKN. Research by the UCT Group has indicated a division shown diagrammatically in Fig 2.3.

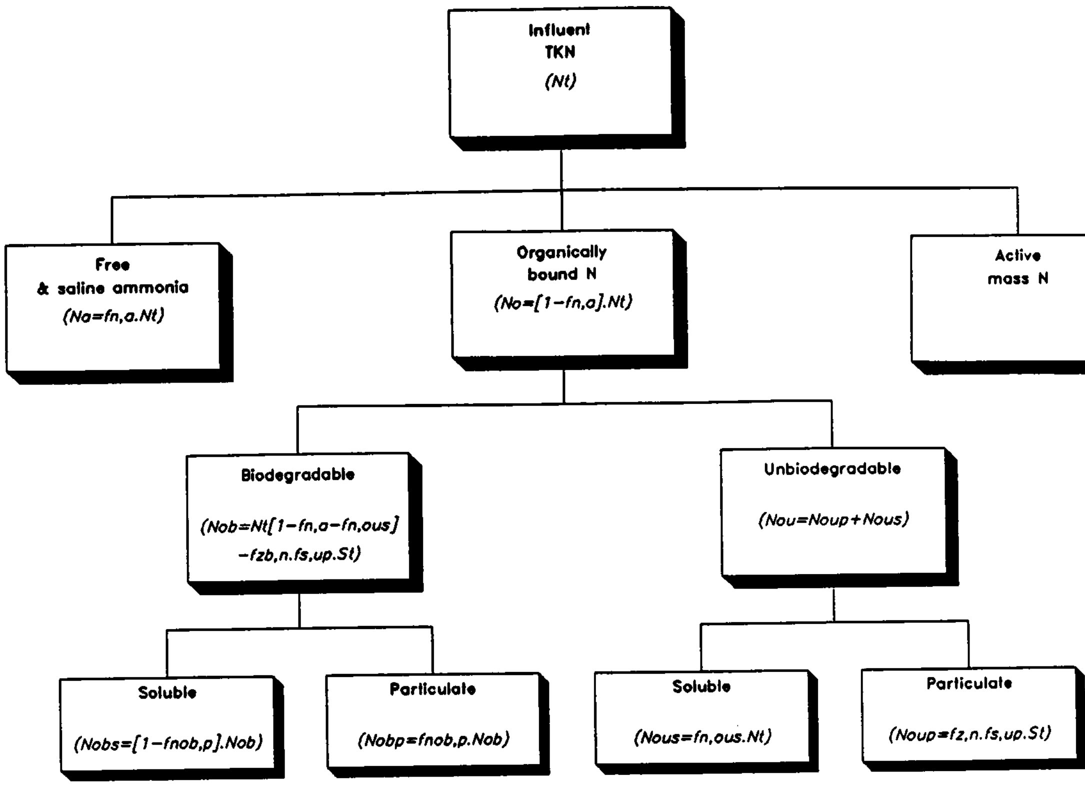


Fig 2.3: Division of the influent TKN into its constituent fractions.

Free and saline ammonia and organically bound fractions

The first subdivision of the influent TKN (N_{ti}) is into free and saline ammonia (N_{ai}) and organically bound N (N_{oi}). The influent organically bound nitrogen (N_{oi}) is divided further into two sub-fractions, unbiodegradable (N_{oui}) and biodegradable (N_{obi}).

Unbiodegradable organic nitrogen: The influent unbiodegradable organic nitrogen (N_{oui}) has two forms, unbiodegradable **soluble** (N_{ousi}) and unbiodegradable **particulate** (N_{oupi}). The unbiodegradable organic nitrogen forms are hypothesized to be unaffected by biological action in the system. The soluble unbiodegradable organic nitrogen passes out in the secondary settling tank overflow. The particulate unbiodegradable organic nitrogen is associated with the unbiodegradable particulate COD in the influent and leaves the system via the sludge wastage stream.

Biodegradable organic nitrogen: The influent biodegradable organic nitrogen (N_{obi}) is in two forms, soluble (N_{obsi}) and particulate (N_{obpi}).

Quantifying the division

Again, from practical considerations the various nitrogen components of the influent wastewater TKN are defined by the fractional constants, $f_{N,a}$, $f_{N,ous}$ and $f_{Nob,p}$:

$f_{N,a}$ = fraction of the **total** influent TKN which is free and saline ammonia;

$f_{N,ous}$ = fraction of the **total** influent TKN which is organic unbiodegradable soluble;

$f_{Nob,p}$ = fraction of the **organic biodegradable** N which is particulate.

Using the fractional constants the division of the influent total TKN (N_{ti}) into the various components may be expressed as follows:

$$N_{ai} = f_{N,a} \cdot N_{ti} \tag{2.6}$$

$$N_{ousi} = f_{N,ous} \cdot N_{ti} \tag{2.7}$$

The influent unbiodegradable particulate organic nitrogen (N_{oupi}) is associated with the influent unbiodegradable particulate COD (S_{upi}), and is expressed as follows:

$$N_{oupi} = f_{Z,N} \cdot f_{S,up} \cdot S_{ti} \tag{2.8}$$

where $f_{Z,N}$ = fraction of VSS (COD units) which is N.

The influent biodegradable organic nitrogen (N_{obi}) is given by the difference of the total influent TKN (N_{ti}) and the sum of N_{ai}, N_{ousi} and N_{oupi}:

$$\begin{aligned} N_{obi} &= N_{ti} - N_{ai} - N_{ousi} - N_{oupi} \\ &= N_{ti}\,(1 - f_{N,a} - f_{N,ous}) - f_{Z,N} \cdot f_{S,up} \cdot S_{ti} \end{aligned} \tag{2.9}$$

The soluble (N_{obsi}) and particulate (N_{obpi}) influent biodegradable organic nitrogen concentrations are respectively:

$$N_{obsi} = (1 - f_{Nob,p})\, N_{obi} \tag{2.10}$$

$$N_{obpi} = f_{Nob,p} \cdot N_{obi} \tag{2.11}$$

To specify the wastewater fully, values must be assigned to the total influent TKN concentration (N_{ti}) and the fractions $f_{N,a}$, $f_{N,ous}$ and $f_{Nob,p}$. The magnitudes of the fractions will vary from influent to influent, and will differ for an unsettled and a primary settled wastewater. In the program default values are provided for raw and settled wastewaters; these are approximate average values observed for municipal wastewaters in South Africa. Procedures for the determination of these values are set out in **Chapter 6**.

Following the IAWPRC recommendations, an influent biological (active) mass nitrogen fraction (N_{ZBi}) is included in both models,

$$N_{ZBi} = f_{ZB,N} \cdot S_{ZBi}$$

where $f_{ZB,N}$ = fraction of biological (active) mass which is nitrogen.

Quantification of this nitrogen fraction depends on the influent biological (active) mass, which can be determined only by trial and error simulations, see earlier.

CHAPTER 3

INSTALLING THE PROGRAMS

THE SUPPLIED DISKS

Three 5¼ inch floppy disks will be found on the inside of the front cover of the manual. Two of the floppy disks, labelled 'UCTOLD Listed Version' and 'IAWPRC Listed Version', contain listings of the two computer programs; the programs are written using Borland's Turbo Pascal version 4.0 and can be listed with the Borland compiler/editor. Also, alterations to the programs can be made using this compiler/editor. [*Should the user make any alterations, the authors request that details of these alterations be forwarded to them for possible incorporation in future versions of the programs.*]

The third floppy disk, labelled 'Distribution disk' contains compiled versions of the two computer programs, **UCTOLD** and **IAWPRC**. This chapter contains instructions on the installation and use of the two activated sludge system simulation programs supplied on the distribution disk.

IMPORTANT NOTE !!

No attempt should be made to run the programs directly from the distribution disk. The instructions in this Chapter should be followed to set up the programs. Users are advised, for their own protection, to make a back-up copy of the distribution disk and to store the original. Instructions on how to make a back-up disk follow in the section below.

BACK-UP COPIES

The distribution disk is formatted for a standard 5¼ inch disk, 360K disk drive, and can be read by an IBM PC or compatible. The following procedure can be used to make a back-up copy of the distribution disk:

- Find a new (or unused) floppy disk.

- Boot up your computer.

- At the system prompt type **diskcopy A: B:** and press **<Enter>**. The message **Insert source diskette into drive A:** will be displayed on the screen. Remove any disk from drive A and replace it with the distribution disk.

- If your system has two floppy disk drives, the screen will say **Insert destination diskette into drive B:**. In that case remove any disk from drive B, replacing it with the blank disk. If your system has only one floppy drive then you will be swapping disks in drive A. Remember that the distribution disk is the **source** disk and the blank disk is the **destination** disk.

- Now press **<Enter>**. The computer will start reading from the source disk in drive A.

- If you have a two-drive system the computer will then write to the destination disk in drive B and continue reading from drive A and writing to drive B until copying is complete. If you have a one-drive system you will be asked to alternatively put the destination disk and the source disk in drive A until copying is finished.

FILES ON THE DISTRIBUTION DISK

The distribution disk contains the following four files:

UCTOLD.EXE
IAWPRC.EXE
SQ555.DAT
RETRIEVE.EXE

The files UCTOLD.EXE and IAWPRC.EXE are the two executable program files. SQ555.DAT is a data file containing a diurnal influent pattern. This data set can be used as an example. RETRIEVE.EXE is the program used to convert data files generated by the simulation programs into a format which can be imported into a spreadsheet package (see **Appendix A**).

SYSTEM REQUIREMENTS

The programs will run on an IBM PC or compatible machine using DOS 3.0 or greater. The amount of RAM memory in the machine will determine the size of system which can be simulated using the programs. With 256K of memory, systems with only one or two reactors can be simulated. The maximum number of reactors increases to twelve with 640K of memory. When using the programs the maximum allowable number of reactors is determined automatically by the program, depending on the amount of memory detected in the particular computer. It is possible that the maximum allowable number of reactors will be reduced if memory resident utilities, such as menu operating programs, have been loaded.

The programs will detect whether or not a graphics adapter is resident in the computer and select the appropriate screen driver automatically. The graphics adapters which can be detected are CGA, EGA, VGA, ATT, PC3270 and Hercules. If a colour card is resident then the programs output will appear in various colours on the screen. In the section of the programs where dynamic system response is simulated one of the options is to display results graphically. If the computer does not have graphics capabilities then a message to this effect is displayed on the screen should this option be selected.

When graphs are displayed on the screen the user may elect to obtain a printed copy of the graph. Graphs can be printed on a 9-pin Epson-compatible dot matrix printer with graphics capabilities; a range of Epson-compatible dot matrix printers has been used to produce graphical output successfully.

The programs use space in 'high' memory to store data. It is possible that problems may be encountered when executing the programs if memory-resident utilities such as spelling checkers have been loaded into memory.

The speed of execution of the simulation programs is dependent on the type of computer used. As a guideline the following execution times were measured when running the program **IAWPRC** on a 10-MHz IBM AT compatible, for a two-reactor contact stabilization system subjected to a square wave influent loading pattern.

> **Steady state solution:** 18 seconds.
>
> **Dynamic solution:** 10 minutes to simulate the performance over five 24-hour cycles, i.e. 1 minute per cycle per reactor.

This example is a particularly 'difficult' one from a numerical point of view for two reasons; (1) the difference in reactor sizes demands a large number of iterations to converge to the steady state solution and necessitates small integration steps, and (2) the sudden changes in the influent pattern compounds the problem of small integration steps. Execution times should be considerably shorter for problems with more standard reactor configurations and less discontinuous influent patterns, and for more advanced computers.

SETTING UP ON A FLOPPY DISK SYSTEM

The two programs, **UCTOLD** and **IAWPRC**, take up most of the space on the distribution disk - approximately 330K of the available 360K. This does not leave much space for storing data files generated by the user. It is therefore suggested that separate floppy disks be set up for *each* program. The procedure outlined below should be followed:

For the **UCTOLD** program:

- Place the distribution disk in drive A.
- Place a blank formatted disk labelled **UCTOLD** in drive B.
- At the DOS prompt on the screen (probably **A:>** or **B:>**) type the following instructions (pressing **<Enter>** at the end of each line):

  copy a:uctold.exe b:
  copy a:retrieve.exe b:
  copy a:sq555.dat b

For the **IAWPRC** program:

- Place the distribution disk in drive A.
- Place a blank formatted disk labelled **IAWPRC** in drive B.

- At the DOS prompt on the screen (probably **A:> or B:>**) type the following instructions (pressing **<Enter>** at the end of each line):

  ```
  copy a:iawprc.exe b:
  copy a:retrieve.exe b:
  copy a:sq555.dat b:
  ```

SETTING UP ON A HARD DISK SYSTEM

The most convenient method for running the programs on a hard disk system is to execute each program from its own subdirectory. Typically these would be called UCTOLD and IAWPRC. The procedure outlined below would then be followed:

- Assuming that the hard disk is designated as drive C, type the following commands (pressing **<Enter>** at the end of each line):

  ```
  c:
  cd c:\
  mkdir uctold
  mkdir iawprc
  ```

  If the hard disk is not designated as drive C, replace C by the relevant designation code.

- Place the distribution disk in the disk drive (usually drive A).

- Change to the subdirectory **UCTOLD** by entering the command

  ```
  cd c:\uctold
  ```

- At the DOS prompt on the screen (**C:>** or **C:\UCTOLD>**) type the following instructions (pressing **<Enter>** at the end of each line):

  ```
  copy a:uctold.exe
  copy a:retrieve.exe
  copy a:sq555.dat
  ```

- Change to the subdirectory **IAWPRC** by entering the command

  ```
  cd c:\iawprc
  ```

- At the DOS prompt on the screen (**C:>** or **C:\IAWPRC>**) type the following instructions (pressing **<Enter>** at the end of each line:

  ```
  copy a:iawprc.exe
  copy a:retrieve.exe
  copy a:sq555.dat
  ```

CHAPTER 4

RUNNING THE PROGRAMS

STARTING PROGRAM EXECUTION

In **Chapter 3**, procedures were described to set up the programs for execution on a floppy drive and a hard drive system. In this Chapter execution of the programs is described for the **UCTOLD** program. If the **IAWPRC** program is to be run, then substitute **IAWPRC** for **UCTOLD**.

Floppy disk system

- Boot up the computer from a DOS disk in drive A:
- Remove the DOS disk from drive A: and replace with disk labelled **UCTOLD** (see **Chapter 3**)
- To initiate execution of the program, at the DOS prompt **A:>** the user must type

  UCTOLD

  and then press the carriage return key (**<RETURN>**, **<C/R>**, **<ENTER>** key on different keyboards).
- The program will be loaded into memory from disk, after which the title page will appear on the screen as shown below.

```
UCTOLD

ACTIVATED SLUDGE SYSTEM SIMULATION PROGRAM - DECEMBER 1990

COPYRIGHT - University of Cape Town - All Rights Reserved Worldwide

Disk may be copied but not offered for lease or sale
in part or in full.

Depts Chemical/Civil Eng.        Program written by P Dold,
University of Cape Town            A Billing, M Wentzel,
Rondebosch 7700  South Africa      G Ekama and G Marais.

                                        Hit key to continue...
```

Hard disk system

- Boot up the system from the hard disk

- At the DOS prompt C:> change to the subdirectory **UCTOLD** (see **Chapter 3**) by entering the command

  cd c:\uctold

- To initiate execution of the program, at the DOS prompt (either C:> or C:\UCTOLD>) type

  UCTOLD

  and press the carriage return key

- The program will be loaded into memory from the hard disk, after which the title page will appear on the screen as shown above.

Once the title page appears on the screen any key may be pressed to continue execution of the program

INTERACTIVE DATA INPUT BY THE USER

The activated sludge simulation program is 'menu driven'. That is, the user selects a desired program option from a list of possible options displayed on the screen. In this program an option is selected by moving a shaded block (or 'highlight) on the screen to cover the appropriate option and then pressing the **<RETURN>** key on the keyboard. (Details of how to move the highlight between options are given later.) This approach should be familiar to users of many commercial software packages and is adopted to simplify program operation and to minimize the amount of typing by the user. Even though the program is 'menu driven' the user nevertheless is required to input certain information from the keyboard such as reactor volumes, COD concentrations, the response to queries (usually Yes/No), and so on. The actions required from the user to input this information should be self-evident. However, to obviate any confusion, a brief instruction on keyboard input follows before program operation is demonstrated.

Keyboard data input typed in by the user falls into three categories:

- **Single characters:** At a number of stages the user must respond to a query printed on the screen. For example, whether or not a set of data must be listed by a printer. In response to these queries a single alphabetical character must be pressed on the keyboard. For example if the query were as follows:

  Print data? Y/N...

  the user merely has to press either the **Y** or **N** key to respond. It is not necessary to press the **<RETURN>** key after typing the letter **Y** or **N**. Furthermore, the program is case-insensitive; that is, either upper- or lower-case letters may be typed (**Y** or **y** / **N** or **n** in this case).

- **Numerical data:** In many instances the user is required to input numerical data; for example, a reactor number or volume. To enter these data the user types in the appropriate value and then presses the **<RETURN>** key. The **Backspace** key may be used to delete typing errors before the **<RETURN>** key is pressed.

- **File names:** Diurnal influent patterns and diurnal response results may be stored on disk for subsequent use. To input a file name the user types in the alphanumeric string and then presses the **<RETURN>** key. The maximum number of characters allowed in a file name is eight. If the user attempts to enter a file name with more than eight characters only the first eight characters will be assigned to the file name. The files containing diurnal influent patterns and diurnal results are automatically given the extension **.DAT** and **.DID**, respectively. When entering a file name it is not necessary to type in the extension.

These instructions apply to data input throughout the program. In the demonstration of program operation which follows the instructions will not be repeated at each point where input by the user is required. For example, the instructions may state that 'the user must enter the COD concentration'. This implies that the user must type in the value and then press the **<RETURN>** key.

The user will find that it is very difficult to upset program operation when inputting information. Protection against typing incorrect keys is built into the program wherever possible. Here are three examples:

- If either a **Y** or an **N** is expected by the program any other key pressed by the user is ignored.

- If a numerical value is to be entered all keys other than those in the set [0,1,...,8,9] (and the decimal point for non-integer values) are ignored if pressed.

- If the user is required to enter a reactor number when there are, say, three reactors in the configuration only the **1**, **2** or **3** keys will be accepted as input.

UNITS FOR REACTOR VOLUMES AND FLOWRATES

Concentration units of g m^{-3} (mg ℓ^{-3}) are used throughout the program. In the case of reactor volumes and flowrates, the user may select one of four possible sets of consistent dimensions. For each set the volumetric quantity is the same (either litres, cubic metres, megalitres or million gallons) and the flowrates are specified as the volumetric quantity per day.

When the title page is visible and any key is pressed, the program advances and the window below will appear on the screen.

```
USER INSTRUCTION ON DIMENSIONS FOR FLOWRATES AND REACTOR VOLUMES

   Any consistent pair of dimensions may be used for entering
   reactor volumes and flowrates. Select one of the following
   pairs:

        VOLUMES    FLOWRATES

   1.      l         l d-1              l  = litres
   2.      m3        m3 d-1             m3 = cubic metres
   3.      Ml        Ml d-1             Ml = megalitres (1000 m3)
   4.      mg        mg d-1 (mgd)       mg = million gallons

        Enter selected dimensions (1,2,3 or 4)...
```

To select the appropriate units the user must enter either **1**, **2**, **3** or **4** as indicated on the screen. The selected units will be used when the program prints information or prompts the user for volumes and flow rates. After the units have been selected, the program automatically advances to the main menu.

THE MAIN MENU

The main program menu is as follows

```
       ******  MAIN  SIMULATION  PROGRAM  MENU  ******
-------------------------------------------------------------
                  INFLUENT DATA
                  PLANT CONFIGURATION
                  OPERATING PARAMETERS
                  STEADY STATE SIMULATION
                  DIURNAL SIMULATION
                  KINETIC CONSTANTS
                  STOICHIOMETRY

                  EXIT FROM PROGRAM
```

```
      Press <Arrows> or <SpaceBar> to move selection

             Press <Return> to select option
```

Eight options, including the option to exit from the program, appear in the upper window of the main menu screen. An option is selected by moving the highlighted block[1] to that option and pressing the **<RETURN>** key. The cursor (-) blinks alongside the highlighted option. The highlighted block may be moved between options by pressing the "up arrow" ($\uparrow$) or "down arrow" ($\downarrow$) keys or by pressing the **spacebar** (this scrolls the highlight downwards). Brief instructions on how to move between options and select an option are shown in the lower window on the screen.

---

[1]In this manual the relevant highlighted option will be indicated by a pointer in the printout of the program screen.

Three options necessarily must be executed before the response of a system can be simulated: **INFLUENT DATA, PLANT CONFIGURATION** and **OPERATING PARAMETERS**. If the user attempts to execute the options **STEADY STATE SIMULATION** or **DIURNAL SIMULATION** before the necessary data have been supplied a message such as that shown below will be flashed on the screen indicating which data are required. The main menu will then appear with the appropriate option highlighted.

```
First specify plant CONFIGURATION....
```

Once the **INFLUENT DATA, PLANT CONFIGURATION** and **OPERATING PARAMETERS** options have been executed the user may select the **STEADY STATE SIMULATION** and then the **DIURNAL SIMULATION** options in that order. In the simulations, default values for the kinetic and stoichiometric constants will be used unless changes have been made to these constants via the **KINETIC CONSTANTS** or **STOICHIOMETRY** options.

The **KINETIC CONSTANTS** option can be selected only if the plant operating temperature has been specified (in the **OPERATING PARAMETERS** option). This is to allow calculation of the values of the constants at the operating temperature based on the values at the reference temperature of 20°C.

In the remainder of this Chapter, operation of the program is demonstrated using the UCTOLD program as an example. Each of the options in the main menu will be covered in turn with a detailed explanation of all data input options, sub-menus and so on. The instructions for each main menu option will commence on a new page for ease of reference.

INFLUENT DATA

The program requires the influent wastewater COD and TKN concentrations and the influent flowrate. There are a number of options for inputting these data:

- If the user wishes to analyze only steady state behaviour then the average influent values may be input directly from the keyboard.
- If the user wishes to analyze diurnal behaviour, then the time-varying influent flow and concentration data must be supplied; the data can be input either from the keyboard or loaded from a disk file. Once the diurnal data have been entered the average influent flowrate and COD and TKN concentrations are computed.
- If the user wishes to analyze steady state behaviour, but the average influent values are calculated from a diurnal pattern, data input is as for the diurnal case.

When the user has selected the set of units for reactor volumes and flowrates the main menu appears with the **INFLUENT DATA** option highlighted as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

        INFLUENT DATA
        PLANT CONFIGURATION
        OPERATING PARAMETERS
        STEADY STATE SIMULATION
        DIURNAL SIMULATION
        KINETIC CONSTANTS
        STOICHIOMETRY

        EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

If the **INFLUENT DATA** option is selected from the main menu a window will appear on the screen as follows

```
**** INFLUENT WASTEWATER CHARACTERISTICS ****

Calculate averages from a Diurnal Pattern? Y/N...
```

If **Y** is entered continue on page 4.10; if **N**, continue overleaf.

Entering steady state influent data

If only steady state behaviour is to be analyzed, and the average influent and concentrations are known, then the user presses the **N** key. The following will appear

```
****  INFLUENT WASTEWATER CHARACTERISTICS  ****

 Average Influent COD (g COD m-3)  =
```

The user is prompted to enter the average influent COD concentration, thereafter the average influent TKN concentration. Once both values have been entered a second window appears on the screen towards the lower right corner.

```
****  INFLUENT WASTEWATER CHARACTERISTICS  ****

 Average Influent COD (g COD m-3)  =  555

 Average Influent TKN (g N m-3)    =  55
```

```
 Settled or Raw sewage? (S/R)....
```

Default values for the characteristics of raw and settled sewage are provided in the program. These are average values for the various COD and TKN fractions encountered with South African municipal wastewaters. The user makes the appropriate selection by entering **S** or **R** for settled or raw sewage, respectively. The screen will appear as follows with the selected values of the fractions as well as the average influent COD and TKN concentrations and a default value for the influent alkalinity appearing in the main window. A legend explaining the meaning of the fractional values is shown in a window at the lower left of the screen.

```
****  WASTEWATER  CHARACTERISTICS  ****

Sti(avg) g COD m-3                    555.000
Nti(avg) g N m-3                       55.000
Fbs      g COD g-1 COD                  0.200
Fs,us    g COD g-1  COD                 0.050
Fs,up    g COD g-1 COD                  0.130
Fn,a     g N g-1 N                      0.750
Fnob,p   g N g-1 N                      0.500
Fn,ous   g N g-1 N                      0.030
Fs,zbh   g Zbh COD g-1 COD              0.000
VSS/TSS  g VSS g-1 TSS                  0.750
Inf Alk  mole m-3                      10.000
                                          RETURN TO MENU
```

```
Fbs     = readily biodegradable/biodegradable COD
Fs,us = frac infl COD that is sol unbiodegrad
Fs,up = frac infl COD that is part unbiodegrad
Fn,a    = frac infl TKN that is NH3/NH4
Fnob,p= frac organic bio N that is part
Fn,ous= frac infl TKN that is organic unbio sol
Fs,zbh= frac infl COD that is heterotrophs
```

```
Hit <Arrows>/<SpaceBar>
  to move selection

 Hit <Return> to enter
    new value for
  selected constant
```

Changes to any of the values (influent COD, TKN and alkalinity as well as the sewage fractions) can be made at this stage. Any updated data will remain current during execution of the program, unless changed again. When the program execution is halted and then is re-run, the original default values will appear.

Instructions for changing any of the data are provided in the window at the lower right of the screen. To update a value the highlighted block must be moved from **RETURN TO MENU** to the value which is to be changed. This is achieved by pressing the 'up arrow' ($\uparrow$) or 'down arrow' ($\downarrow$) keys or by pressing the **spacebar** (this scrolls the highlight from the top downwards). When the highlight appears on the value to be changed the user presses the **<RETURN>** key, and is prompted for a new value as shown below. On typing the new value and pressing the **<RETURN>** key the screen is updated and the highlight reverts to the **RETURN TO MENU** position.

```
****  WASTEWATER  CHARACTERISTICS  ****

Sti(avg) g COD m-3                    555.000
Nti(avg) g N m-3                       55.000
Fbs      g COD g-1 COD                  0.200
Fs,us    g COD g-1  COD                 0.050
Fs,up    g COD g-1 COD                  0.130   New value =
Fn,a     g N g-1 N                      0.750
Fnob,p   g N g-1 N                      0.500
Fn,ous   g N g-1 N                      0.030
Fs,zbh   g Zbh COD g-1 COD              0.000
VSS/TSS  g VSS g-1 TSS                  0.750
Inf Alk  mole m-3                      10.000
                                                RETURN TO MENU
```

```
Fbs     = readily biodegradable/biodegradable COD
Fs,us = frac infl COD that is sol unbiodegrad
Fs,up = frac infl COD that is part unbiodegrad
Fn,a    = frac infl TKN that is NH3/NH4
Fnob,p= frac organic bio N that is part
Fn,ous= frac infl TKN that is organic unbio sol
Fs,zbh= frac infl COD that is heterotrophs
```

```
Hit <Arrows>/<SpaceBar>
  to move selection

 Hit <Return> to enter
    new value for
  selected constant
```

When the user is satisfied with the influent data the <RETURN> key must be pressed with the highlight in the **RETURN TO MENU** position. Before the main menu is re-listed a new window appears at the centre of the screen as shown below.

```
**** WASTEWATER CHARACTERISTICS ****

Sti(avg) g COD m-3          555.000
Nti(avg) g N m-3             55.000
Fbs      g COD g-1 COD        0.200
Fs,us    g COD g-1 COD        0.050
Fs,up    g COD g-1 COD        0.130
Fn,a     g N g-1 N            0.750

Do you want a hardcopy of wastewater characteristics? Y/N...

Fbs
Fs,us = frac infl COD that is sol unbiodegrad
Fs,up = frac infl COD that is part unbiodegrad
Fn,a  = frac infl TKN that is NH3/NH4
Fnob,p= frac organic bio N that is part
Fn,ous= frac infl TKN that is organic unbio sol
Fs,zbh= frac infl COD that is heterotrophs

Bar>
to move selection

Hit <Return> to enter
new value for
selected constant
```

When a printer is attached to the computer the user may respond by entering **Y**; else **N** is entered to proceed. If **Y** is entered there will be a short delay (of a few seconds) while data are sent to the printer. At a later stage, directly after simulating either the steady state or diurnal response, it is again possible to obtain a printout of the sewage characteristics. After entering **Y** or **N** the program reverts to the main menu and the highlight appears on the **PLANT CONFIGURATION** option as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

         INFLUENT DATA
         PLANT CONFIGURATION
         OPERATING PARAMETERS
         STEADY STATE SIMULATION
         DIURNAL SIMULATION
         KINETIC CONSTANTS
         STOICHIOMETRY

         EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

Instructions for entering **PLANT CONFIGURATION** data commence on Page 4.21.

Entering a diurnal influent pattern

When the **INFLUENT DATA** option is first selected from the main menu the user is required to specify whether or not a diurnal influent pattern is required, as shown in the following window.

```
****  INFLUENT WASTEWATER CHARACTERISTICS  ****

Calculate averages from a Diurnal Pattern? Y/N...
```

The procedure to follow when **N** is entered was explained in the previous section Entering steady state influent data. If the Y key is pressed the following window will appear.

```
*****  DIURNAL DATA INPUT FROM :  *****

DISK
KEYBOARD

RETURN TO MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

If one or more diurnal influent patterns have already been stored on disk one of these may be retrieved by selecting the **DISK** option. This option is outlined in the next section Retrieving a diurnal pattern from disk on Page 4.16.

If diurnal data are to be input for the first time or if a new diurnal pattern is to be input then the **KEYBOARD** option must be selected. The following instruction on the format of the diurnal data then appears on the screen.

```
        CREATING DATA FOR DYNAMIC ANALYSIS

To utilise the program for the simulation of dynamic process
response it is necessary first to set up arrays which contain
the time-varying values of the influent FLOWRATE, COD and TKN.
The program requires 12 sets of values; that is, two-hourly
values. These values can be stored on the working disk drive
once the user is satisfied.

        ** Hit any key to continue...
```

Program execution is advanced by pressing any key as instructed. A new screen appears with a main window and an input window as shown below. The user is prompted to input twelve sets of data (influent flowrate and concentrations of COD and TKN) as requested in the input window at the lower right of the screen. These are the average values for each of the twelve two-hour intervals, starting with the first interval at 0h00. Each time interval extends from the given time for the next two hours. For example, data for time 0h00 applies for the period 0h00 to 2h00.

```
        ******  DIURNAL  INPUT  PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)

                              Time (hours) =  0h00 :

                                Flowrate ( m3 d-1) =
```

As each set of flowrate, COD and TKN is completed, the set appears in the main window. For example, when six sets have been input the screen may be as follows.

```
        ******  DIURNAL  INPUT  PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
     1         0.0         0.0          555.0         55.0
     2         2.0         0.0          555.0         55.0
     3         4.0         0.0          555.0         55.0
     4         6.0         0.0          555.0         55.0
     5         8.0        22.0          555.0         55.0
     6        10.0        22.0          555.0         55.0

                              Time (hours) =  12h00 :

                                Flowrate ( m3 d-1) =
```

When all twelve sets of data have been entered the weighted mean values are calculated and appear in the main window at the lower left as shown below. For this case a square wave influent pattern has been set up. The flowrate is 22 $m^3\ d^{-1}$ for a period of 12 hours (6 intervals of 2 hours starting at 08h00 and ending at 20h00; note that record number 10 with time 18.0 applies for the period 18h00 to 20h00) with no flow for the remainder of the diurnal cycle. The concentrations of COD ($gCOD\ m^{-3}$) and TKN ($gN\ m^{-3}$) are intended to be constant at 555 and 55, respectively. However, an error appears in the sixth record where the TKN has been entered incorrectly as 44 $gN\ m^{-3}$.

```
            ******  DIURNAL INPUT PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
    1          0.0          0.0          555.0        55.0
    2          2.0          0.0          555.0        55.0
    3          4.0          0.0          555.0        55.0
    4          6.0          0.0          555.0        55.0
    5          8.0         22.0          555.0        55.0
    6         10.0         22.0          555.0        44.0
    7         12.0         22.0          555.0        55.0
    8         14.0         22.0          555.0        55.0
    9         16.0         22.0          555.0        55.0
   10         18.0         22.0          555.0        55.0
   11         20.0          0.0          555.0        55.0
   12         22.0          0.0          555.0        55.0

  ** Calculated Mean Values :       Change any values? Y/N....
        Flowrate =     11.0
        COD      =    555.0
        TKN      =     53.2
```

After all the data have been entered corrections can be made by entering a **Y** at the prompt in the input window. The record number (6 here) is entered. The user is prompted to enter all three values for that record. The mean influent values are recalculated and the screen is updated. When the data are correct the user advances the program by responding with an **N** to the request for changes. The message in the input window will change as follows.

```
            ******  DIURNAL INPUT PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
    1          0.0          0.0          555.0        55.0
    2          2.0          0.0          555.0        55.0
    3          4.0          0.0          555.0        55.0
    4          6.0          0.0          555.0        55.0
    5          8.0         22.0          555.0        55.0
    6         10.0         22.0          555.0        55.0
    7         12.0         22.0          555.0        55.0
    8         14.0         22.0          555.0        55.0
    9         16.0         22.0          555.0        55.0
   10         18.0         22.0          555.0        55.0
   11         20.0          0.0          555.0        55.0
   12         22.0          0.0          555.0        55.0

  ** Calculated Mean Values :
        Flowrate =     11.0         Do you wish to store these
        COD      =    555.0               data on disk ?  Y/N....
        TKN      =     53.2
```

The diurnal pattern may be stored on disk for later use. When the user wishes to store the influent pattern **Y** is entered at the prompt and a name is requested as shown below.

```
            ******  DIURNAL INPUT PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
    1          0.0         0.0          555.0        55.0
    2          2.0         0.0          555.0        55.0
    3          4.0         0.0          555.0        55.0
    4          6.0         0.0          555.0        55.0
    5          8.0        22.0          555.0        55.0
    6         10.0        22.0          555.0        55.0
    7         12.0        22.0          555.0        55.0
    8         14.0        22.0          555.0        55.0
    9         16.0        22.0          555.0        55.0
   10         18.0        22.0          555.0        55.0
   11         20.0         0.0          555.0        55.0
   12         22.0         0.0          555.0        55.0

  ** Calculated Mean Values :      Enter data file name :
       Flowrate =    11.0
       COD      =   555.0                 e.g. INFDATA1
       TKN      =    55.0
                                          FileName =
```

Once a name has been entered (maximum of 8 characters) the data are stored on disk. The data are stored in a file of the specified name with the extension **.DAT** on the disk (and in the directory) from where the program was executed. Once the data have been stored a window appears at the centre of the screen as shown below.

```
            ******  DIURNAL INPUT PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
    1          0.0         0.0          555.0        55.0
    2          2.0         0.0          555.0        55.0
    3          4.0         0.0          555.0        55.0
    4          6.0         0.0          555.0        55.0

  Do you want a hardcopy of diurnal input? Y/N...

  ** Calculated Mean Values :      Enter data file name :
       Flowrate =    11.0
       COD      =   555.0                 e.g. INFDATA1
       TKN      =    55.0
                                          FileName = sq555
```

The user is prompted to enter **Y** to obtain a printout of the diurnal influent pattern. After the **Y** or **N** key is pressed the screen will appear as follows.

```
****** DIURNAL INPUT PATTERN ******

Record No   Time (h)   Flow( m3 d-1)   COD (g m-3)   TKN (g m-3)
    1         0.0          0.0           555.0          55.0
    2         2.0          0.0           555.0          55.0
    3         4.0          0.0           555.0          55.0
    4         6.0          0.0           555.0          55.0

Do you want a hardcopy of diurnal input? Y/N...

** Calculated Mean Values :
      Flowrate =    11.0
      COD      =   555.0
      TKN      =    55.0

Settled or Raw sewage? (S/R)....
```

The user must specify whether raw or settled sewage characteristics are to be selected. Once the **R** or the **S** key is pressed the list of wastewater characteristics will appear on the screen as shown below. The values for the average influent COD and TKN concentration are those calculated from the diurnal influent pattern, and the values for the constants are default values obtained for South African municipal wastewaters.

```
**** WASTEWATER CHARACTERISTICS ****

Sti(avg) g COD m-3                 555.000
Nti(avg) g N m-3                    55.000
Fbs      g COD g-1 COD               0.200
Fs,us    g COD g-1  COD              0.050
Fs,up    g COD g-1 COD               0.130
Fn,a     g N g-1 N                   0.750
Fnob,p   g N g-1 N                   0.500
Fn,ous   g N g-1 N                   0.030
Fs,zbh   g Zbh COD g-1 COD           0.000
VSS/TSS  g VSS g-1 TSS               0.750
Inf Alk  mole m-3                   10.000
                                        RETURN TO MENU

Fbs     = readily biodegradable/biodegradable COD
Fs,us = frac infl COD that is sol unbiodegrad
Fs,up = frac infl COD that is part unbiodegrad
Fn,a  = frac infl TKN that is NH3/NH4
Fnob,p= frac organic bio N that is part
Fn,ous= frac infl TKN that is organic unbio sol
Fs,zbh= frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
   to move selection

 Hit <Return> to enter
     new value for
   selected constant
```

Changes can be made to the default values where appropriate by following the instructions in the lower right window. This procedure has been outlined in the previous section <u>**Entering steady state influent data**</u>. When the user is satisfied with the influent data the **<RETURN>** key must be pressed with the highlight in the **RETURN TO MENU** position. Before the main menu is re-listed a new window appears at the centre of the screen as shown overleaf.

```
**** WASTEWATER CHARACTERISTICS ****

Sti(avg) g COD m-3                 555.000
Nti(avg) g N m-3                    55.000
Fbs      g COD g-1 COD               0.200
Fs,us    g COD g-1  COD              0.050
Fs,up    g COD g-1 COD               0.130
Fn,a     g N g-1 N                   0.750

  Do you want a hardcopy of wastewater characteristics? Y/N...

Fbs                                              Bar>
Fs,us = frac infl COD that is sol unbiodegrad     to move selection
Fs,up = frac infl COD that is part unbiodegrad
Fn,a  = frac infl TKN that is NH3/NH4            Hit <Return> to enter
Fnob,p= frac organic bio N that is part            new value for
Fn,ous= frac infl TKN that is organic unbio sol   selected constant
Fs,zbh= frac infl COD that is heterotrophs
```

When a printer is attached to the computer the user may respond by entering **Y**; else an **N** is entered to proceed. If a **Y** is entered there will be a short delay (of a few seconds) while data are sent to the printer. At a later stage, directly after simulating either the steady state or diurnal response, it is again possible to obtain a printout of the sewage characteristics. After entering **Y** or **N** the program returns to the main menu with the highlight appearing on the **PLANT CONFIGURATION** option as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

          INFLUENT DATA
          PLANT CONFIGURATION
          OPERATING PARAMETERS
          STEADY STATE SIMULATION
          DIURNAL SIMULATION
          KINETIC CONSTANTS
          STOICHIOMETRY

          EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

Instructions for entering **PLANT CONFIGURATION** data commence on page 4.21.

Retrieving a diurnal pattern from disk

When the **INFLUENT DATA** option is first selected from the main menu the user is required to specify whether or not a diurnal influent pattern is required in the following window.

```
**** INFLUENT WASTEWATER CHARACTERISTICS ****

Calculate averages from a Diurnal Pattern? Y/N...
```

The procedure to follow when an **N** is entered was explained in the earlier section **Entering steady state influent data**. If the **Y** key is pressed the following window will appear.

```
***** DIURNAL DATA INPUT FROM : *****

DISK
KEYBOARD

RETURN TO MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

If diurnal data are to be input for the first time or if a new diurnal pattern is to be input then the **KEYBOARD** option must be selected. The procedure to follow in this case is outlined in the previous section **Entering a diurnal influent Pattern**.

If one or more diurnal influent patterns have already been stored on disk one of these may be retrieved by selecting the **DISK** option. As an example, the following may appear on the screen.

```
EXISTING FILES

SQ555.DAT
```

```
Enter data file name:
     e.g. INFLNT1

FileName =
```

The names of all the stored influent data files are listed in three columns in the main window. The user is prompted to enter the name of the file to be retrieved. It is not necessary to type the extension .DAT.

The diurnal data are displayed in the same format as in the previous section <u>Entering a diurnal influent pattern</u>. An example is shown below. Changes to the data can be made, and the modified data stored if desired, as described in that section. Thereafter the user must specify whether raw or settled sewage characteristics are to be selected, and changes made to the default values as appropriate according to the procedure outlined in the section <u>Entering steady state influent data</u>. Finally, the user can obtain a printout if required before returning to the main menu. The procedure to follow once a data file has been retrieved is very similar to that presented in the previous section. However, the procedure is now outlined for the sake of completeness.

****** DIURNAL INPUT PATTERN ******

Record No	Time (h)	Flow(m3 d-1)	COD (g m-3)	TKN (g m-3)
1	0.0	0.0	555.0	55.0
2	2.0	0.0	555.0	55.0
3	4.0	0.0	555.0	55.0
4	6.0	0.0	555.0	55.0
5	8.0	22.0	555.0	55.0
6	10.0	22.0	555.0	55.0
7	12.0	22.0	555.0	55.0
8	14.0	22.0	555.0	55.0
9	16.0	22.0	555.0	55.0
10	18.0	22.0	555.0	55.0
11	20.0	0.0	555.0	55.0
12	22.0	0.0	555.0	55.0

```
** Calculated Mean Values :
     Flowrate =    11.0
     COD      =   555.0
     TKN      =    55.0
```

```
Change any values? Y/N....
```

Changes can be made by entering a **Y** at the prompt in the input window. The record number is entered and the user is prompted to enter all three values for that record. The mean influent values are recalculated and the screen is updated. When no more changes are required, an **N** is entered and a window appears at the centre of the screen as shown below.

```
        ******  DIURNAL INPUT PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
    1          0.0         0.0           555.0        55.0
    2          2.0         0.0           555.0        55.0
    3          4.0         0.0           555.0        55.0
    4          6.0         0.0           555.0        55.0

  Do you want a hardcopy of diurnal input? Y/N...

  ** Calculated Mean Values :          Change any values? Y/N....
        Flowrate =     11.0
        COD      =    555.0
        TKN      =     55.0
```

The user is prompted to enter a **Y** to obtain a printout of the diurnal influent pattern. After the **Y** or **N** key is pressed the screen will appear as follows.

```
        ******  DIURNAL INPUT PATTERN  ******

Record No    Time (h)  Flow( m3 d-1)  COD (g m-3)  TKN (g m-3)
    1          0.0         0.0           555.0        55.0
    2          2.0         0.0           555.0        55.0
    3          4.0         0.0           555.0        55.0
    4          6.0         0.0           555.0        55.0

  Do you want a hardcopy of diurnal input? Y/N...

  ** Calculated Mean Values :
        Flowrate =     11.0
        COD      =    555.0            Settled or Raw sewage? (S/R)....
        TKN      =     55.0
```

The user must specify whether raw or settled sewage characteristics are to be selected. Once the **R** or the **S** key is pressed the list of wastewater characteristics will appear on the screen as shown below. The values for the average influent COD and TKN concentration are those calculated from the diurnal influent pattern and the values of the constants are default values obtained for South African municipal wastewaters.

```
**** WASTEWATER CHARACTERISTICS ****

Sti(avg) g COD m-3              555.000
Nti(avg) g N m-3                 55.000
Fbs      g COD g-1 COD            0.200
Fs,us    g COD g-1  COD           0.050
Fs,up    g COD g-1 COD            0.130
Fn,a     g N g-1 N                0.750
Fnob,p   g N g-1 N                0.500
Fn,ous   g N g-1 N                0.030
Fs,zbh   g Zbh COD g-1 COD        0.000
VSS/TSS  g VSS g-1 TSS            0.750
Inf Alk  mole m-3                10.000
                                      RETURN TO MENU

Fbs    = readily biodegradable/biodegradable COD
Fs,us  = frac infl COD that is sol unbiodegrad
Fs,up  = frac infl COD that is part unbiodegrad
Fn,a   = frac infl TKN that is NH3/NH4
Fnob,p= frac organic bio N that is part
Fn,ous= frac infl TKN that is organic unbio sol
Fs,zbh= frac infl COD that is heterotrophs

Hit <Arrows>/<SpaceBar>
to move selection

Hit <Return> to enter
new value for
selected constant
```

Changes can be made to the default values as appropriate by following the instructions in the lower right window. This procedure has been outlined in the previous section Entering steady state influent data. When the user is satisfied with the influent data the **RETURN TO MENU** option is selected. Before the main menu is re-listed a new window appears at the centre of the screen as shown below.

```
**** WASTEWATER CHARACTERISTICS ****

Sti(avg) g COD m-3              555.000
Nti(avg) g N m-3                 55.000
Fbs      g COD g-1 COD            0.200
Fs,us    g COD g-1  COD           0.050
Fs,up    g COD g-1 COD            0.130
Fn,a     g N g-1 N                0.750

Do you want a hardcopy of wastewater characteristics? Y/N...

Fbs                                            Bar>
Fs,us = frac infl COD that is sol unbiodegrad    to move selection
Fs,up = frac infl COD that is part unbiodegrad
Fn,a  = frac infl TKN that is NH3/NH4            Hit <Return> to enter
Fnob,p= frac organic bio N that is part          new value for
Fn,ous= frac infl TKN that is organic unbio sol  selected constant
Fs,zbh= frac infl COD that is heterotrophs
```

When a printer is attached to the computer the user may respond by entering **Y**; else an **N** is entered to proceed. If a **Y** is entered there will be a short delay (of a few seconds) while data are sent to the printer. At a later stage, directly after simulating either the steady state or diurnal response, it is again possible to obtain a printout of the sewage characteristics. After entering **Y** or **N** the program returns to the main menu with the highlight appearing on the **PLANT CONFIGURATION** option as shown overleaf.

****** MAIN SIMULATION PROGRAM MENU ******

INFLUENT DATA
PLANT CONFIGURATION
OPERATING PARAMETERS
STEADY STATE SIMULATION
DIURNAL SIMULATION
KINETIC CONSTANTS
STOICHIOMETRY

EXIT FROM PROGRAM

Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option

PLANT CONFIGURATION

When the user has completed the input of influent data the main menu appears with the **PLANT CONFIGURATION** option highlighted.

```
****** MAIN SIMULATION PROGRAM MENU ******

INFLUENT DATA
PLANT CONFIGURATION
OPERATING PARAMETERS
STEADY STATE SIMULATION
DIURNAL SIMULATION
KINETIC CONSTANTS
STOICHIOMETRY

EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

On pressing the **<RETURN>** key to select the **PLANT CONFIGURATION** option the following screen appears.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors
```

```
Number of reactors (max 12)  =
```

In the input window at the lower right of the screen the user is prompted to enter the number of completely mixed reactors in the system; the maximum allowable number is indicated and has been calculated by the program from the amount of memory available. Thereafter input from the user is dependent on whether one or more reactors are specified. For multiple reactor configurations continue on Page 4.25.

<u>Single reactor configuration</u>

Once the number of reactors has been specified (1 here) the user is prompted to input the reactor volume as shown below. The units of volume (selected earlier) appear in the input window as a reminder.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 1

     Gp 2. Reactor Vols,m3

No. 1:
```

```
Enter reactor volume(s) - m3:

     Reactor No. 1  =
```

When the volume of the reactor has been entered the screen will appear as shown below. In an activated sludge system with only one reactor all of the influent necessarily must enter this reactor and the reactor must be aerated. A default dissolved oxygen (DO) concentration of 3 gO m^{-3} (mgO ℓ^{-1}) is assumed.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 1

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction    Gp 4. Aeration/DO
No. 1:            7.00                  1.00                   3.0
```

```
Do you wish to change any parameters?

       Y/N....
```

The user may change the data by following the instructions in the input window. If a change is to be made and the Y key is pressed the input window will appear as follows.

```
**** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 1

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:           7.00                1.00                 3.0
```

```
Do you wish to change any parameters?

    In which group?  1,2,..,4
```

The user must specify in which group of data the change is required. For example, if a change is desired in the DO concentration the 4 key is pressed, and the user is prompted for a new DO value in the input window as follows.

```
**** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 1

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:           7.00                1.00                 3.0
```

```
Aerated reactor DO setpoints:
     e.g.  3    (g O m-3)

  Reactor No. 1 : DO setpoint =
```

When no more changes are required to the data and an **N** is entered at the prompt a window will appear at the centre of the screen as follows.

```
**** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 1

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:          7.00                  1.00                  2.0

Do you want a hardcopy of process configuration? Y/N...
```

If a printer is attached to the computer the user may respond by pressing the **Y** key to obtain a printout of the configuration data; else an **N** is entered. At a later stage, directly after simulating either the steady state or diurnal response, there is another opportunity to print the configuration data. After entering **Y** or **N** the program returns to the main menu with the highlight on the **OPERATING PARAMETERS** option as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

          INFLUENT DATA
          PLANT CONFIGURATION
          OPERATING PARAMETERS
          STEADY STATE SIMULATION
          DIURNAL SIMULATION
          KINETIC CONSTANTS
          STOICHIOMETRY

          EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

Multiple reactor configuration

If there is more than one reactor in the system the user may wish to specify that the feed is split between reactors, that certain reactors are unaerated, and that mixed liquor recycles are included. The convention for numbering reactors is that the last reactor is the one preceding the settling tank, and flow is sequential from reactor to reactor by number; that is, from Reactor 1 to 2 to 3 and so on. Examples of reactor numbering for different configurations are illustrated in Appendix C.

When two or more reactors are specified the user is prompted to input the reactor volumes sequentially. For example, if two reactors have been specified and the first reactor volume has been entered as 2 m^3 the screen will appear as follows.

```
 **** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 2

    Gp 2. Reactor Vols,m3

No. 1:          2.00
No. 2:
```

```
Enter reactor volume(s) - m3:

     Reactor No. 2  =
```

Once all the reactor volumes have been specified (2 and 5 m^3 in this case) the screen will be similar to that shown below.

```
 **** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 2

    Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction

No. 1:          2.00
No. 2:          5.00
```

```
Fractional influent distribution:

  Fraction to Reactor No. 1 =
```

The influent flow may be split between reactors. The user is prompted to enter the fraction fed to each reactor sequentially. Input continues until either the sum of the fractions equals one or the faction to the second last reactor is specified (the remainder is assumed to enter the last reactor).

When the feed distribution has been entered the screen may appear as follows.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 2

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:           2.00               1.00               Aerated
No. 2:           5.00                                  Aerated
```

```
Reactor aeration pattern:

     Make any changes?  Y/N.....
```

Initially it is assumed that all reactors are aerated. To change the status of a reactor between aerated and unaerated the user enters a Y at the prompt. The user is then prompted to enter the number of the reactor to change in the input window as shown below.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 2

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:           2.00               1.00               Aerated
No. 2:           5.00                                  Aerated
```

```
Reactor aeration pattern:

        Change Reactor No.
```

The screen is updated to reflect the change in aeration pattern. This procedure is repeated until the user is satisfied with the aeration pattern. When no more changes are required the N key is pressed and the user is prompted to enter the DO concentration in the aerated reactors sequentially. For example, in the two reactor case if the first reactor is unaerated, the following will appear.

```
**** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 2

    Gp 2. Reactor Vols,m3     Gp 3. Feed Fraction    Gp 4. Aeration/DO

No. 1:           2.00               1.00              Unaerated
No. 2:           5.00                                 Aerated
```

```
Aerated reactor DO setpoints:
      e.g.  3   (g O m-3)

  Reactor No. 2 : DO setpoint =
```

When all the DO concentrations have been input the screen appears as shown below and the user specifies whether or not there are any mixed liquor recycles between reactors. A maximum of two recycles (A and B) other than the return activated sludge (RAS) recycle may be specified.

```
**** PROCESS CONFIGURATION ****      Gp 1. Number of Reactors = 2

    Gp 2. Reactor Vols,m3     Gp 3. Feed Fraction    Gp 4. Aeration/DO

No. 1:           2.00               1.00              Unaerated
No. 2:           5.00                                    3.0

Gp 5. Recycles:   Use to include/remove mixed liquor recycles.
```

```
Any inter-reactor mixed liquor
    recycles?  Y/N....
```

If the Y key is pressed details for the A recycle are requested. The user first enters the number of the reactor from which the recycle is taken (as shown below), and then the number of the reactor to which the recycle is pumped.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 2

    Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:            2.00               1.00               Unaerated
No. 2:            5.00                                     3.0

Gp 5. Recycles:   Use to include/remove mixed liquor recycles.
```

```
A recycle : Out of Reactor No.
```

When this information has been supplied the screen is updated and the user specifies whether or not a second recycle is required as shown below.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 2

    Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:            2.00               1.00               Unaerated
No. 2:            5.00                                     3.0

Gp 5. Recycles:   Use to include/remove mixed liquor recycles.

 A recycle : Out of Reactor No.2
             Into Reactor No.1
```

```
Another mixed liquor recycle?

       Y/N....
```

If a Y is pressed data for the B recycle are requested in the same format as for the A recycle.

If any mixed liquor recycles have been specified the user is prompted to input the number of the reactor to which the RAS recycle is directed in the screen shown below. Had no recycles been specified the program would automatically assume that the RAS recycle is pumped to the first reactor.

```
**** PROCESS CONFIGURATION ****       Gp 1. Number of Reactors = 2

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:           2.00               1.00               Unaerated
No. 2:           5.00                                     3.0

Gp 5. Recycles:    Use to include/remove mixed liquor recycles.

 A recycle : Out of Reactor No.2
             Into Reactor No.1
```

```
RAS recycle from settler to

    Reactor No.
```

This completes the data input for **PLANT CONFIGURATION**. The user can make changes to the data by entering a **Y** at the prompt as shown below. The procedure for making changes has been outlined for the single reactor case earlier.

```
**** PROCESS CONFIGURATION ****       Gp 1. Number of Reactors = 2

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction   Gp 4. Aeration/DO

No. 1:           2.00               1.00               Unaerated
No. 2:           5.00                                     3.0

Gp 5. Recycles:    Use to include/remove mixed liquor recycles.

 A recycle : Out of Reactor No.2
             Into Reactor No.1

 RAS recycle to Reactor No.1
```

```
Do you wish to change any parameters?

       Y/N....
```

When no more changes are required to the data and an **N** is entered at the prompt a window will appear at the centre of the screen as follows.

```
**** PROCESS CONFIGURATION ****        Gp 1. Number of Reactors = 2

     Gp 2. Reactor Vols,m3    Gp 3. Feed Fraction    Gp 4. Aeration/DO

No. 1:          2.00                1.00                 Unaerated
No. 2:          5.00                                        3.0

Gp
 A

 R  Do you want a hardcopy of process configuration? Y/N...
```

If a printer is attached to the computer the user may respond by pressing the **Y** key to obtain a printout of the configuration data; else an **N** is entered. At a later stage, directly after simulating either the steady state or diurnal response, there is another opportunity to print the configuration data. After entering **Y** or **N** the program returns to the main menu with the highlight on the **OPERATING PARAMETERS** option as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

          INFLUENT DATA
          PLANT CONFIGURATION
          OPERATING PARAMETERS
          STEADY STATE SIMULATION
          DIURNAL SIMULATION
          KINETIC CONSTANTS
          STOICHIOMETRY

          EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

        Press <Return> to select option
```

OPERATING PARAMETERS

Once the system configuration data have been entered the main menu reappears with the **OPERATING PARAMETERS** option highlighted.

```
****** MAIN  SIMULATION  PROGRAM  MENU  ******

          INFLUENT DATA
          PLANT CONFIGURATION
          OPERATING PARAMETERS
          STEADY STATE SIMULATION
          DIURNAL SIMULATION
          KINETIC CONSTANTS
          STOICHIOMETRY

          EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

        Press <Return> to select option
```

On pressing the **<RETURN>** key to select the **OPERATING PARAMETERS** option the following screen appears.

```
****** PLANT OPERATING PARAMETERS ******
```

```
Operating sludge age (SRT) of plant (days) =
```

The user is prompted to enter the sludge age (SRT) of the plant in the input window at the lower right of the screen as shown above. The sludge age appears in the main window and the user is prompted to input the temperature within the plant (not ambient) in °C as shown overleaf.

```
****** PLANT OPERATING PARAMETERS ******
1   SRT (total)            d        =      5.0
```

```
Operating temperature of plant (degC) =
```

If a diurnal influent pattern has been set up in the **INFLUENT DATA** option the average influent flowrate will appear automatically in the main window immediately after the temperature is entered, as shown below. The user is then prompted to enter the RAS recycle flowrate in the input window. However, if average influent COD and TKN concentrations were entered in the **INFLUENT DATA** option then the user is prompted to enter the average influent flowrate before the RAS recycle flowrate.

```
****** PLANT OPERATING PARAMETERS ******
1   SRT (total)            d        =      5.0
2   Process Temperature    degC     =     22.0

Flow rates:
3   Influent flow          m3 d-1   =     11.0
```

```
RAS-recycle Flow (m3 d-1) =
```

If the mixed liquor recycles have been specified in the **PLANT CONFIGURATION** option the user is prompted to enter the appropriate flow rates. The mixed liquor recycle flowrates are assumed to be constant irrespective of any variation in the influent flowrate. For the two reactor example discussed earlier the screen would appear as follows.

```
****** PLANT OPERATING PARAMETERS ******

1   SRT (total)            d        =    5.0
2   Process Temperature    degC     =   22.0

Flow rates:
3   Influent flow          m3 d-1   =   11.0
4   RAS recycle flow       m3 d-1   =   11.0
```

```
A recycle Flow (m3 d-1) =
```

This completes the input of data for the **OPERATING PARAMETERS** option. The user may make changes to the data by following the instructions in the input window as shown below.

```
****** PLANT OPERATING PARAMETERS ******

1   SRT (total)            d        =    5.0
2   Process Temperature    degC     =   22.0

Flow rates:
3   Influent flow          m3 d-1   =   11.0
4   RAS recycle flow       m3 d-1   =   11.0
5   A-recycle flow         m3 d-1   =   33.0
```

```
Do you wish to change any parameters?

        Y/N....
```

If a change is to be made and the **Y** key is pressed the user is prompted to specify in which group of data the change is required, as follows.

```
****** PLANT OPERATING PARAMETERS ******

1   SRT (total)              d          =     5.0
2   Process Temperature      degC       =    22.0

Flow rates:
3   Influent flow            m3 d-1     =    11.0
4   RAS recycle flow         m3 d-1     =    11.0
5   A-recycle flow           m3 d-1     =    33.0
```

```
Do you wish to change any parameters?

    In which group?  1,2,..,5
```

For example, if a change is desired in the RAS recycle rate the 4 key is pressed, and the user is prompted for a new flowrate value in the input window as shown below.

```
****** PLANT OPERATING PARAMETERS ******

1   SRT (total)              d          =     5.0
2   Process Temperature      degC       =    22.0

Flow rates:
3   Influent flow            m3 d-1     =    11.0
4   RAS recycle flow         m3 d-1     =    11.0
5   A-recycle flow           m3 d-1     =    33.0
```

```
RAS-recycle Flow (m3 d-1) =
```

The main window is updated to reflect the new value and the user is prompted for more changes. When no more changes are required the N key is pressed at the prompt and a new window appears at the centre of the screen as follows.

```
****** PLANT OPERATING PARAMETERS ******

1   SRT {total}            d        =     5.0
2   Process Temperature    degC     =    22.0

Flow rates:
3   Influent flow          m3 d-1   =    11.0
4
5
```

```
Do you want a hardcopy of operating conditions? Y/N...
```

If a printer is attached to the computer a printout of the operating parameters may be obtained by entering a **Y** ; else an **N** is entered to proceed. At a later stage, directly after simulating either the steady state or diurnal response, there is another opportunity to obtain a printout of the operating parameters. After entering **Y** or **N** the program reverts to the main menu with the highlight on the **STEADY STATE SIMULATION** option as shown below.

```
****** MAIN  SIMULATION  PROGRAM  MENU ******

          INFLUENT DATA
          PLANT CONFIGURATION
          OPERATING PARAMETERS
          STEADY STATE SIMULATION
          DIURNAL SIMULATION
          KINETIC CONSTANTS
          STOICHIOMETRY

          EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

         Press <Return> to select option
```

STEADY STATE SIMULATION

Once the **INFLUENT DATA, PLANT CONFIGURATION** and **OPERATING PARAMETERS** options have been executed the main menu reappears with the **STEADY STATE SIMULATION** option highlighted as shown below. The user may press the **<RETURN>** key to initiate the steady state calculation. In this case default values for the kinetic and stoichiometric constants are used in the simulation. Under certain circumstances the user may wish to make changes to the kinetic and stoichiometric constants prior to the steady state calculation via the main menu **KINETIC CONSTANTS** or **STOICHIOMETRY** options, see later.

```
****** MAIN SIMULATION PROGRAM MENU ******

          INFLUENT DATA
          PLANT CONFIGURATION
          OPERATING PARAMETERS
          STEADY STATE SIMULATION
          DIURNAL SIMULATION
          KINETIC CONSTANTS
          STOICHIOMETRY

          EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

When the **STEADY STATE SIMULATION** option is selected the program first calculates the mixed liquor wastage rate required for the selected reactor volumes and specified sludge age. The sludge wastage stream is assumed to be withdrawn from the last reactor in the series; that is, the reactor preceding the settling tank. If the solids concentration is approximately constant from reactor to reactor, calculation of the wastage rate is a trivial problem. However, if there are appreciable differences in solids concentration between reactors an iterative calculation procedure must be employed. These differences occur in systems where the influent does not enter the first reactor (such as the contact stabilization process) or where the RAS recycle is not pumped to the first reactor (such as the UCT process). To cover all possibilities the iterative procedure always is used in the program. During the calculation a line of data relating to the solids distribution is printed on the screen at each iteration as shown below. These data are of no concern to the user, but provide an indication that calculation is continuing.

```
Iterative Wastage Rate Calculation                  Wastage rate

785.71  785.71                                           1.400


                   Wastage rate =     1.400
```

When the calculation is completed the required wastage rate is displayed on the screen for a few seconds. There is no need for the user to note this value because it can be printed, if required, together with the operating parameters after completing the steady state calculation.

Calculation of the steady state solution commences once the wastage rate has been determined. Because calculation of the steady state response involves solution of a set of nonlinear algebraic equations an iterative procedure is utilized. Starting from an empirical estimate of the solution the estimates are improved until the solution is reached. During calculation, the concentrations of eight compounds are printed on the screen at each iteration together with an error factor, MaxF. When the system comprises more than one reactor, five of the concentration values are for selected compounds in the first reactor and three for compounds in the last reactor, as shown in the example below.

First Reactor					Last Reactor			
Zbh	Zba	Sads	Sbs	No3	Sbs	Na	No3	MaxF
1039.38	34.93	103.94	18.20	0.00	1.00	0.50	38.92	1.000E+10
1020.84	31.10	120.62	0.18	0.02	0.01	1.21	6.79	1.487E+03
1024.43	31.67	81.49	0.99	0.17	0.01	1.74	6.82	7.411E+02
1022.74	31.62	87.84	0.01	0.53	0.00	1.87	7.17	9.772E+02
1021.51	31.61	89.68	0.41	1.52	0.00	1.87	8.15	1.899E+02
1019.97	31.59	92.97	0.38	2.78	0.00	1.87	9.41	3.456E+01
1019.68	31.59	93.60	0.39	3.01	0.00	1.87	9.65	3.194E-01

If there is only one reactor in the system the eight parameters are compound concentrations from that reactor. During the solution procedure each of the concentrations converges to its steady state value. The solution is accepted as satisfactory when the error factor, MaxF, in the right-hand-most column decreases to less than 10^{-3}.

When the steady state solution is found the results appear on the screen. An example of the results format is shown below for a two reactor system where the first reactor is unaerated and where the **UCTOLD** model has been used to simulate system behaviour. In the first column the model compounds are listed together with seven additional parameters of interest. These are the volatile and total settleable solids concentrations; oxygen utilization rates for heterotrophs, autotrophs (nitrifiers), and the total; the denitrification rate; and the TKN concentration. The influent concentrations of the model compounds are listed under the title **INPUT**. The values of all parameters in each reactor are listed below the corresponding reactor number under the title **REACTOR**. Units for each parameter are shown at the right-hand side.

```
COMPOUND                 INPUT         REACTOR
                                     1        2
Zbh (hetero.)        =     0.0  1019.7  1043.7 g COD m-3
Zba (autotrophs)     =     0.0    31.6    32.5 g COD m-3
Ze (endog.)          =     0.0   267.8   272.8 g COD m-3
Zi (prt unb COD)     =    72.1   566.9   566.9 g COD m-3
Sads (adsorb.COD)    =     0.0    93.6    31.8 g COD m-3
Senm (enmesh COD)    =   364.1    14.1     4.1 g COD m-3
Nobp (prt bio N)     =     3.6     3.6     2.0 g N m-3
Sbs (sol bio COD)    =    91.0     0.4     0.0 g COD m-3
Na (ammonia N)       =    41.3     9.4     1.9 g N m-3
Nobs (sol org N)     =     3.6     1.1     1.6 g N m-3
No3 (nitrate N)      =     0.0     3.0     9.6 g N m-3
Alkalinity           =    10.0     7.5     6.5 mole m-3
Sus (sol unb COD)    =    27.7    27.7    27.7 g COD m-3

Volatile SS          =          1347.1  1318.7 g VSS m-3
Total SS             =          1796.1  1758.3 g TSS m-3
OUR heterotrophs     =             0.0    19.8 g O2 m-3 h-1
OUR autotrophs       =             0.0    13.5 g O2 m-3 h-1
OUR total            =             0.0    33.3 g O2 m-3 h-1
Denit. rate          =             5.4     0.0 g NO3-N m-3 h-1
TKN                  =            12.1     5.1 g N m-3

*** Hit any key to continue...
```

When the results have been viewed any key may be pressed to advance the program. A window appears at the centre of the screen as shown below.

```
COMPOUND                 INPUT          REACTOR
                                      1      2
Zbh (hetero.)        =     0.0 1019.7 1043.7 g COD m-3
Zba (autotrophs)     =     0.0   31.6   32.5 g COD m-3
Ze (endog.)          =     0.0  267.8  272.8 g COD m-3
Zi (prt unb COD)     =    72.1  566.9  566.9 g COD m-3
Sads (adsorb.COD)    =     0.0   93.6   31.8 g COD m-3
Senm (enmesh COD)    =   364.1   14.1    4.1 g COD m-3
Nobp (prt bio N)     =     3.6    3.6    2.0 g N m-3
Sbs (so
Na (amm
Nobs (s
No3 (ni
Alkalin  Do you want a hardcopy of steady state results ? Y/N...
Sus (so

Volatil
Total S
OUR heterotrophs     =            0.0   19.8 g O2 m-3 h-1
OUR autotrophs       =            0.0   13.5 g O2 m-3 h-1
OUR total            =            0.0   33.3 g O2 m-3 h-1
Denit. rate          =            5.4    0.0 g NO3-N m-3 h-1
TKN                  =           12.1    5.1 g N m-3

*** Hit any key to continue...
```

If a printer is attached to the computer the user may obtain a printout of the steady state results by entering a **Y** at the prompt; else an **N** is entered. After the **Y** or **N** key has been pressed the user may then elect to print one or more of the following sets of data:

- wastewater characteristics
- plant configuration data
- operating data
- kinetic constants
- stoichiometric constants

When the user has responded to the requests for printed output the main menu appears with the **DIURNAL SIMULATION** option highlighted as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

            INFLUENT DATA
            PLANT CONFIGURATION
            OPERATING PARAMETERS
            STEADY STATE SIMULATION
            DIURNAL SIMULATION
            KINETIC CONSTANTS
            STOICHIOMETRY

            EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

DIURNAL SIMULATION

When the **DIURNAL SIMULATION** option is selected from the main menu a window will appear with a sub-menu as shown below.

```
***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

Eight options, including one to return to the main menu, appear in the upper window of the diurnal sub-menu screen. An option is selected in the same manner as for the main menu by moving the highlighted block to that option and pressing the **<RETURN>** key. The cursor (–) blinks alongside the highlighted option. The highlighted block may be moved between options by pressing the 'up arrow' (↑) or 'down arrow' (↓) keys or by pressing the **spacebar** (this scrolls the highlight downwards). Brief instructions on how to move between options and select an option are shown in the lower window on the screen.

Three options necessarily can be selected only once the **SIMULATE** option has been executed: **HARDCOPY OF RESULTS, GRAPHICAL OUTPUT** and **STORE DATA ON DISK**. If the user attempts to select one of these options before the **SIMULATE** option is executed the message shown below will be flashed on the screen indicating that the **SIMULATE** option should first be executed. The menu will then reappear with the **SIMULATE** option highlighted.

```
First SIMULATE...
```

In the remainder of this section each of the diurnal sub-menu options is demonstrated. Each of the options will be covered in turn with a detailed explanation of all data input options, any sub-menus and so on. The instructions for each diurnal simulation menu option will commence on a new page for ease of reference.

Simulate

When the **SIMULATE** option is selected the program commences computing the response of the system under the diurnal influent pattern. It is assumed that the diurnal pattern is repeated continuously. In this situation the true solution is one where the response of the system is repeated identically from 24-hour cycle to 24-hour cycle. That is, the concentration of each compound (in a given reactor) is the same at a given time from cycle to cycle. Also the concentration of each compound (in a given reactor) will be the same at the start and end of a cycle (0h00 and 24h00). Because the concentrations of the compounds at the start of the cycle are not known before commencing the calculation it is necessary to estimate these. The initial estimates are taken as the steady state solution. Starting from these values the response of the system is simulated by integrating forward in time. The process continues for cycle after cycle until the values of certain parameters are close to equal at the start and end of a cycle. When the differences in the magnitudes of these 'end' values are within specified limits the response is accepted as being sufficiently close to the true solution.

The screen is cleared when the **SIMULATE** option is first selected. An integration time monitor appears towards the base of the screen as shown below.

```
Integration time =   1.25
```

The magnitude displayed by the time counter shows the current position of computation in the cycle and increases from 0.0 to 24.0. The function of the timer is merely as an indicator to the user. The rate at which the counter is incremented is not necessarily constant. This is because a variable steplength algorithm is used for the integration.

Some intermediate results are displayed on the screen when the end of the first 24-hour cycle is reached as shown in the example below. These are the values of ten parameters in the last reactor of the configuration at two-hourly intervals as calculated during the previous cycle. Control of which ten parameters are displayed is discussed in the section <u>Check/change screen/printed output</u> later. The integration time in the current cycle appears at the base of the screen.

***** DIURNAL RESPONSE RESULTS *****

Reactor 2 Cycle number 1

TIME	Zbh	Zba	Sads	Senm	VSS	Sbs	Na	No3	TKN	Ot
0.0	1043.7	32.5	31.8	4.1	1318.7	0.0	1.9	9.6	5.1	33.3
2.0	1040.0	32.7	14.5	3.4	1306.4	-0.0	0.6	10.5	3.8	20.2
4.0	1021.6	32.8	10.3	3.4	1294.2	-0.0	0.4	10.9	3.5	15.6
6.0	1000.7	32.9	9.6	3.3	1282.7	-0.0	0.3	11.2	3.3	14.1
8.0	979.9	33.0	9.4	3.4	1271.5	-0.0	0.3	11.3	3.2	13.4
10.0	970.7	32.7	42.7	5.1	1286.7	0.0	2.9	9.7	5.8	37.1
12.0	977.6	32.3	69.5	5.4	1305.9	0.0	4.6	8.5	7.6	42.2
14.0	988.6	32.0	89.4	5.5	1323.2	0.0	5.5	8.0	8.6	44.4
16.0	1001.2	31.7	105.2	5.6	1339.0	0.0	6.1	7.8	9.3	45.6
18.0	1015.1	31.4	117.7	5.6	1353.5	0.0	6.6	7.8	9.8	46.5
20.0	1029.7	31.2	127.5	5.6	1366.9	0.0	6.9	7.7	10.1	47.2
22.0	1059.2	31.6	67.4	3.6	1346.9	-0.0	1.8	10.5	5.3	38.5
24.0	1068.8	31.8	20.8	3.4	1324.8	-0.0	0.6	11.1	4.4	23.6

Integration time = 1.25

At the end of each cycle the most recent results are displayed in the format shown above. If the computation is progressing satisfactorily the difference between the values of each parameter at the start and end of the cycle should get smaller from cycle to cycle. When the difference is judged by the program to be sufficiently small the results of the current cycle are accepted as the diurnal solution and integration is terminated. An example of the screen output at this point is shown below.

***** DIURNAL RESPONSE RESULTS *****

Reactor 2 Cycle number 7

TIME	Zbh	Zba	Sads	Senm	VSS	Sbs	Na	No3	TKN	Ot
0.0	1084.9	30.1	20.0	3:4	1331.5	-0.0	0.8	11.5	4.6	24.5
2.0	1068.1	30.4	11.2	3.4	1317.5	-0.0	0.6	12.4	3.9	18.3
4.0	1046.5	30.6	10.1	3.4	1305.4	-0.0	0.4	13.0	3.4	15.3
6.0	1024.8	30.7	9.8	3.4	1293.7	-0.0	0.3	13.3	3.3	14.2
8.0	1003.5	30.7	9.5	3.4	1282.2	-0.0	0.3	13.4	3.3	13.7
10.0	993.4	30.4	42.6	5.1	1296.8	0.0	3.1	11.2	6.0	36.6
12.0	999.9	30.1	68.4	5.3	1315.2	0.0	5.0	9.3	8.0	41.6
14.0	1010.6	29.9	86.8	5.4	1331.5	0.0	6.1	8.1	9.2	43.8
16.0	1022.6	29.6	102.0	5.4	1346.7	0.0	6.9	7.6	10.0	45.1
18.0	1035.1	29.4	114.9	5.4	1360.7	0.0	7.4	7.4	10.6	46.0
20.0	1048.4	29.1	125.1	5.5	1373.7	0.0	7.8	7.3	11.0	46.8
22.0	1077.3	29.6	65.1	3.6	1353.5	-0.0	2.7	10.2	6.3	39.9
24.0	1085.5	30.0	20.0	3.4	1331.7	-0.0	0.9	11.5	4.7	24.7

*** Dynamic solution reached...Hit any key to continue...

If a key is pressed the program returns to the diurnal sub-menu with the **HARDCOPY OF RESULTS** option highlighted as shown below.

***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU

Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option

Hard copy of results

When the **HARDCOPY OF RESULTS** option is selected from the diurnal sub-menu the window shown below appears on the screen.

```
Do you want a hardcopy of diurnal results ? Y/N...
```

If a printer is attached to the computer the user may press the **Y** key to obtain a printout of some of the diurnal results; else an **N** is entered. The results which are printed are the values, for each reactor, of ten selected parameters at intervals over the 24-hour cycle. An example of a portion of the output *for one of the reactors* is shown below.

***** DIURNAL RESPONSE RESULTS *****

Reactor 2 Cycle number 7

TIME	Zbh	Zba	Sads	Senm	VSS	Sbs	Na	No3	TKN	Ot
0.0	1084.9	30.1	20.0	3.4	1331.5	-0.0	0.8	11.5	4.6	24.5
0.3	1083.6	30.2	17.6	3.4	1329.4	-0.0	0.8	11.6	4.6	23.4
0.5	1082.0	30.2	15.8	3.4	1327.5	-0.0	0.8	11.7	4.5	22.4
0.8	1080.1	30.2	14.5	3.3	1325.7	-0.0	0.8	11.8	4.4	21.5
1.0	1077.9	30.3	13.5	3.3	1324.0	-0.0	0.8	11.9	4.3	20.8
1.3	1075.6	30.3	12.6	3.4	1322.3	-0.0	0.8	12.1	4.2	20.1
1.5	1073.2	30.3	12.0	3.4	1320.7	-0.0	0.7	12.2	4.1	19.4
1.8	1070.7	30.4	11.5	3.4	1319.1	-0.0	0.7	12.3	4.0	18.8
2.0	1068.1	30.4	11.2	3.4	1317.5	-0.0	0.6	12.4	3.9	18.3
2.3	1065.5	30.4	10.9	3.4	1316.0	-0.0	0.6	12.5	3.8	17.8
2.5	1062.8	30.5	10.7	3.4	1314.4	-0.0	0.6	12.6	3.7	17.3
2.8	1060.1	30.5	10.6	3.3	1312.9	-0.0	0.5	12.7	3.7	16.9
3.0	1057.4	30.5	10.5	3.3	1311.4	-0.0	0.5	12.8	3.6	16.5
3.3	1054.7	30.5	10.3	3.3	1309.9	-0.0	0.5	12.9	3.6	16.2
3.5	1052.0	30.5	10.2	3.4	1308.4	-0.0	0.4	12.9	3.5	15.9
3.8	1049.2	30.6	10.1	3.4	1306.9	-0.0	0.4	13.0	3.5	15.6
4.0	1046.5	30.6	10.1	3.4	1305.4	-0.0	0.4	13.0	3.4	15.3
4.3	1043.8	30.6	10.0	3.4	1303.9	-0.0	0.4	13.1	3.4	15.1
4.5	1041.0	30.6	10.0	3.3	1302.5	-0.0	0.4	13.1	3.4	14.9
4.8	1038.3	30.6	10.0	3.3	1301.0	-0.0	0.3	13.1	3.4	14.8
5.0	1035.6	30.6	10.0	3.3	1299.5	-0.0	0.3	13.2	3.3	14.7
5.3	1032.9	30.6	9.9	3.3	1298.1	-0.0	0.3	13.2	3.3	14.5
5.5	1030.2	30.6	9.8	3.4	1296.6	-0.0	0.3	13.2	3.3	14.4
5.8	1027.5	30.7	9.8	3.4	1295.1	-0.0	0.3	13.2	3.3	14.3
6.0	1024.8	30.7	9.8	3.4	1293.7	-0.0	0.3	13.3	3.3	14.2
6.3	1022.1	30.7	9.7	3.4	1292.2	-0.0	0.3	13.3	3.3	14.1
6.5	1019.5	30.7	9.7	3.4	1290.8	-0.0	0.3	13.3	3.3	14.0
6.8	1016.8	30.7	9.7	3.4	1289.4	-0.0	0.3	13.3	3.3	14.0
7.0	1014.1	30.7	9.7	3.3	1287.9	-0.0	0.3	13.3	3.3	13.9
7.3	1011.5	30.7	9.7	3.3	1286.5	-0.0	0.3	13.4	3.3	13.9
7.5	1008.8	30.7	9.7	3.3	1285.1	-0.0	0.3	13.4	3.3	13.8
7.8	1006.2	30.7	9.6	3.3	1283.6	-0.0	0.3	13.4	3.3	13.8
8.0	1003.5	30.7	9.5	3.4	1282.2	-0.0	0.3	13.4	3.3	13.7
8.2	1000.4	30.7	11.2	4.2	1282.0	0.0	0.5	13.2	3.5	17.0
8.5	997.7	30.7	15.2	4.7	1283.1	0.0	0.8	13.0	3.8	22.0
8.8	995.8	30.7	20.0	4.8	1285.0	0.0	1.2	12.7	4.2	26.2
9.0	994.5	30.6	24.9	4.9	1287.1	0.0	1.6	12.4	4.6	29.4
9.2	993.7	30.6	29.7	4.9	1289.5	0.0	2.0	12.1	5.0	31.9
9.5	993.4	30.5	34.2	5.0	1291.9	0.0	2.4	11.8	5.4	33.8
9.7	993.3	30.5	38.5	5.0	1294.4	0.0	2.7	11.5	5.7	35.3
10.0	993.4	30.4	42.6	5.1	1296.8	0.0	3.1	11.2	6.0	36.6
10.2	993.8	30.4	46.5	5.1	1299.3	0.0	3.4	10.9	6.4	37.6
10.5	994.3	30.4	50.1	5.1	1301.7	0.0	3.6	10.6	6.6	38.5
10.7	995.0	30.3	53.6	5.2	1304.0	0.0	3.9	10.4	6.9	39.2
11.0	995.8	30.3	56.8	5.2	1306.3	0.0	4.1	10.1	7.2	39.8
11.3	996.7	30.2	59.9	5.2	1308.6	0.0	4.4	9.9	7.4	40.4
11.5	997.7	30.2	62.9	5.2	1310.8	0.0	4.6	9.7	7.6	40.8
11.7	998.7	30.2	65.7	5.2	1313.0	0.0	4.8	9.5	7.8	41.3
12.0	999.9	30.1	68.4	5.3	1315.2	0.0	5.0	9.3	8.0	41.6

The selection of the ten parameters which appear in the table of results can be modified via the **CHECK/CHANGE SCREEN/PRINTED OUTPUT** option. The procedure for modifying the selection is discussed later.

In the example above the results are printed at intervals of 15 minutes; for illustrative purposes, only the first 12 hours of the 24 hour cycle are shown that is, there are 49 sets of values. The printing interval is set equal to the *data interval* which appears in the **CHECK/CHANGE INTEGRATION PARAMETERS** sub-menu discussed later. At this point it suffices to note that intervals of 10, 15 or 30 minutes can be selected. The default value is 15 minutes.

The printing of results may take several minutes depending on printer speed, data interval and the number of reactors. When the program is ready to continue after the last set of data has been sent to the printer the diurnal sub-menu will appear on the screen with the highlight on the **HARDCOPY OF RESULTS** option as shown below.

```
***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

Graphical output

When the **GRAPHICAL OUTPUT** option is selected from the diurnal sub-menu the following sub-menu appears on the screen. The user may select between plotting graphical results on the screen (and possibly on a printer) and changing the graph plotting scales.

```
*** DIURNAL RESPONSE GRAPHICAL OUTPUT ***

PLOT GRAPHS
CHECK/CHANGE PLOTTING SCALES

RETURN TO DIURNAL MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

If the **PLOT GRAPHS** option is selected three windows appear on the screen as shown below. The upper right window indicates the variable to be plotted **versus** time over the 24-hour cycle; plots of this variable in each reactor will be obtained. A list of eighteen possible plotting variables appears in a window at the left. The number of variables in this window differs for **IAWPRC** and **UCTOLD.**

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)      g COD m-3
 2 Zba (autotrophs)   g COD m-3
 3 Ze (endog.)        g COD m-3
 4 Zi (prt unb COD)   g COD m-3
 5 Sads (adsorb.COD)  g COD m-3
 6 Senm (enmesh COD)  g COD m-3
 7 Nobp (prt bio N)   g N m-3
 8 Sbs (sol bio COD)  g COD m-3
 9 Na (ammonia N)     g N m-3
10 Nobs (sol org N)   g N m-3
11 No3 (nitrate N)    g N m-3
12 Alkalinity         mole m-3
13 Sus (sol unb COD)  g COD m-3
14 OUR carb     g O2 m-3 h-1
15 OUR nitr     g O2 m-3 h-1
16 OUR tot      g O2 m-3 h-1
17 Volatile solids    g VSS m-3
18 TKN                g N m-3
```

```
SELECTED PLOTTING OUTPUTS

Ot in all reactors
```

```
Change this selection? Y/N...
```

If the user wishes to change the variable to be plotted, a **Y** is entered at the prompt and the user must respond to the query in the lower right window, as follows.

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)        g COD m-3
 2 Zba (autotrophs)     g COD m-3
 3 Ze (endog.)          g COD m-3
 4 Zi (prt unb COD)     g COD m-3
 5 Sads (adsorb.COD)    g COD m-3
 6 Senm (enmesh COD)    g COD m-3
 7 Nobp (prt bio N)     g N m-3
 8 Sbs (sol bio COD)    g COD m-3
 9 Na (ammonia N)       g N m-3
10 Nobs (sol org N)     g N m-3
11 No3 (nitrate N)      g N m-3
12 Alkalinity           mole m-3
13 Sus (sol unb COD)    g COD m-3
14 OUR carb       g O2 m-3 h-1
15 OUR nitr       g O2 m-3 h-1
16 OUR tot        g O2 m-3 h-1
17 Volatile solids      g VSS m-3
18 TKN                  g N m-3
```

```
SELECTED PLOTTING OUTPUTS

Ot in all reactors
```

```
Change this selection? Y/N...

Change to Parameter No.
```

If the user wishes to view a plot of, say, the ammonia concentration instead of the total oxygen utilization rate then a 9 would be entered at the prompt. The selected option will change in the upper right window as follows.

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)        g COD m-3
 2 Zba (autotrophs)     g COD m-3
 3 Ze (endog.)          g COD m-3
 4 Zi (prt unb COD)     g COD m-3
 5 Sads (adsorb.COD)    g COD m-3
 6 Senm (enmesh COD)    g COD m-3
 7 Nobp (prt bio N)     g N m-3
 8 Sbs (sol bio COD)    g COD m-3
 9 Na (ammonia N)       g N m-3
10 Nobs (sol org N)     g N m-3
11 No3 (nitrate N)      g N m-3
12 Alkalinity           mole m-3
13 Sus (sol unb COD)    g COD m-3
14 OUR carb       g O2 m-3 h-1
15 OUR nitr       g O2 m-3 h-1
16 OUR tot        g O2 m-3 h-1
17 Volatile solids      g VSS m-3
18 TKN                  g N m-3
```

```
SELECTED PLOTTING OUTPUTS

Na in all reactors
```

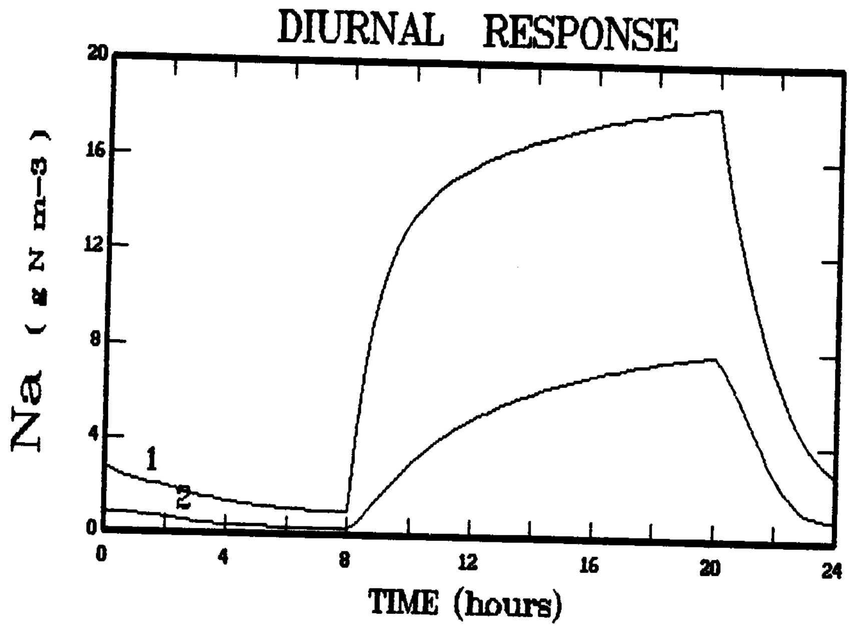
```
Change this selection? Y/N...
```

Once the desired plotting parameter has been selected the user enters an **N** at the prompt in the lower right window. If the computer does not have graphics facilities the message shown below is flashed on the screen. Thereafter the program will return to the diurnal sub-menu.

```
Graphics error

You probably have no graphics card!
```

If the computer has graphics facilities a plot of the response of the selected parameter over the 24-hour cycle will appear on the screen. An example plot is shown below.



Once the user has viewed the plot a copy of the graph may be generated on a printer by entering a **Y** at the prompt. This option should be used only when a 9-pin Epson-compatible dot matrix printer is being used. Once a **Y** or an **N** has been entered the diurnal menu will reappear with the **GRAPHICAL OUTPUT** option highlighted.

```
***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

If the **CHECK/CHANGE PLOTTING SCALES** option is selected from the sub-menu of the **GRAPHICAL OUTPUT** option shown above, changes can be made to default values of the maximum plotting range for the parameters. Two windows appear on the screen as shown below. The possible plotting variables are listed in a window to the left of the screen together with the current values of the maxima of the graphs for the corresponding variables. The list will differ for **IAWPRC** and **UCTOLD**.

MAXIMUM PLOTTING RANGES

1	Zbh (hetero.)	g COD m-3	4000
2	Zba (autotrophs)	g COD m-3	400
3	Ze (endog.)	g COD m-3	4000
4	Zi (prt unb COD)	g COD m-3	4000
5	Sads (adsorb.COD)	g COD m-3	400
6	Senm (enmesh COD)	g COD m-3	1000
7	Nobp (prt bio N)	g N m-3	100
8	Sbs (sol bio COD)	g COD m-3	20
9	Na (ammonia N)	g N m-3	40
10	Nobs (sol org N)	g N m-3	20
11	No3 (nitrate N)	g N m-3	50
12	Alkalinity	mole m-3	20
13	Sus (sol unb COD)	g COD m-3	100
14	OUR carb	g O2 m-3 h-1	100
15	OUR nitr	g O2 m-3 h-1	100
16	OUR tot	g O2 m-3 h-1	100
17	Volatile solids	g VSS m-3	10000
18	TKN	g N m-3	50

```
Alter any ranges ? Y/N...
```

The user must enter either a **Y** or an **N** at the prompt in the window to the right depending on whether or not a change must be made to any of the maximum plotting values. If a **Y** is entered the window on the right will change as follows.

MAXIMUM PLOTTING RANGES

1	Zbh (hetero.)	g COD m-3	4000
2	Zba (autotrophs)	g COD m-3	400
3	Ze (endog.)	g COD m-3	4000
4	Zi (prt unb COD)	g COD m-3	4000
5	Sads (adsorb.COD)	g COD m-3	400
6	Senm (enmesh COD)	g COD m-3	1000
7	Nobp (prt bio N)	g N m-3	100
8	Sbs (sol bio COD)	g COD m-3	20
9	Na (ammonia N)	g N m-3	40
10	Nobs (sol org N)	g N m-3	20
11	No3 (nitrate N)	g N m-3	50
12	Alkalinity	mole m-3	20
13	Sus (sol unb COD)	g COD m-3	100
14	OUR carb	g O2 m-3 h-1	100
15	OUR nitr	g O2 m-3 h-1	100
16	OUR tot	g O2 m-3 h-1	100
17	Volatile solids	g VSS m-3	10000
18	TKN	g N m-3	50

```
Alter any ranges ? Y/N...

Change Scale No.
```

The number of the parameter for which the scale must change is entered at the prompt. If, say, a **9** is entered (for the parameter ammonia here) the user is prompted to enter a new maximum plotting range for this parameter as shown below.

MAXIMUM PLOTTING RANGES

1	Zbh (hetero.)	g COD m-3	4000
2	Zba (autotrophs)	g COD m-3	400
3	Ze (endog.)	g COD m-3	4000
4	Zi (prt unb COD)	g COD m-3	4000
5	Sads (adsorb.COD)	g COD m-3	400
6	Senm (enmesh COD)	g COD m-3	1000
7	Nobp (prt bio N)	g N m-3	100
8	Sbs (sol bio COD)	g COD m-3	20
9	Na (ammonia N)	g N m-3	40
10	Nobs (sol org N)	g N m-3	20
11	No3 (nitrate N)	g N m-3	50
12	Alkalinity	mole m-3	20
13	Sus (sol unb COD)	g COD m-3	100
14	OUR carb	g O2 m-3 h-1	100
15	OUR nitr	g O2 m-3 h-1	100
16	OUR tot	g O2 m-3 h-1	100
17	Volatile solids	g VSS m-3	10000
18	TKN	g N m-3	50

Alter any ranges ? Y/N...

Change Scale No. 9

New Na max. range =

When the new value is entered the appropriate maximum plotting range is updated in the window to the left. The user is then prompted for more changes to the maximum plotting ranges. If an **N** is entered the graphics sub-menu reappears with the **PLOT GRAPHS** option highlighted as shown below.

*** DIURNAL RESPONSE GRAPHICAL OUTPUT ***

PLOT GRAPHS
CHECK/CHANGE PLOTTING SCALES

RETURN TO DIURNAL MENU

Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option

When the user has finished viewing the graphs the **RETURN TO DIURNAL MENU** option is selected. The program returns to the diurnal sub-menu with the **GRAPHICAL OUTPUT** option highlighted as shown below.

```
***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

The diurnal response patterns shown in the plots are constructed by joining data points at intervals with straight lines. If the intervals are short then the graph will mimic the response closely. The interval between data points is equal to the data interval which appears in the **CHECK/CHANGE INTEGRATION PARAMETERS** *sub-menu discussed later. At this point it suffices to note that intervals of 10, 15 or 30 minutes can be selected. The default value is 15 minutes.*

Check/change screen/printed output

The response of ten parameters is listed either on the screen during computation of the diurnal response (**SIMULATE** option) or by the printer when the option **HARDCOPY OF RESULTS** is selected from the diurnal sub-menu. This is the maximum number that can be listed conveniently on the screen or on 80 column printer paper. The ten selected output variables may be modified via the **CHECK/CHANGE SCREEN/PRINTED OUTPUT** option.

If the **CHECK/CHANGE SCREEN/PRINTED OUTPUT** option is selected from the diurnal sub-menu three windows appear on the screen as shown below. The upper right window lists the ten current output variables. A list of eighteen possible output variables appears in a window at the left. The number of variables in this window differs for **IAWPRC** and **UCTOLD**.

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)         g COD m-3
 2 Zba (autotrophs)      g COD m-3
 3 Ze (endog.)           g COD m-3
 4 Zi (prt unb COD)      g COD m-3
 5 Sads (adsorb.COD)     g COD m-3
 6 Senm (enmesh COD)     g COD m-3
 7 Nobp (prt bio N)      g N m-3
 8 Sbs (sol bio COD)     g COD m-3
 9 Na (ammonia N)        g N m-3
10 Nobs (sol org N)      g N m-3
11 No3 (nitrate N)       g N m-3
12 Alkalinity            mole m-3
13 Sus (sol unb COD)     g COD m-3
14 OUR carb        g O2 m-3 h-1
15 OUR nitr        g O2 m-3 h-1
16 OUR tot         g O2 m-3 h-1
17 Volatile solids       g VSS m-3
18 TKN                   g N m-3
```

```
SELECTED OUTPUTS

 1   Zbh
 2   Zba
 3   Sads
 4   Senm
 5   VSS
 6   Sbs
 7   Na
 8   No3
 9   TKN
10   Ot
```

```
Change any of the above ? Y/N...
```

The selected output variables can be changed one at a time. If a **Y** is entered at the prompt the user must respond to the query in the window to the lower right as shown below.

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)         g COD m-3
 2 Zba (autotrophs)      g COD m-3
 3 Ze (endog.)           g COD m-3
 4 Zi (prt unb COD)      g COD m-3
 5 Sads (adsorb.COD)     g COD m-3
 6 Senm (enmesh COD)     g COD m-3
 7 Nobp (prt bio N)      g N m-3
 8 Sbs (sol bio COD)     g COD m-3
 9 Na (ammonia N)        g N m-3
10 Nobs (sol org N)      g N m-3
11 No3 (nitrate N)       g N m-3
12 Alkalinity            mole m-3
13 Sus (sol unb COD)     g COD m-3
14 OUR carb        g O2 m-3 h-1
15 OUR nitr        g O2 m-3 h-1
16 OUR tot         g O2 m-3 h-1
17 Volatile solids       g VSS m-3
18 TKN                   g N m-3
```

```
SELECTED OUTPUTS

 1   Zbh
 2   Zba
 3   Sads
 4   Senm
 5   VSS
 6   Sbs
 7   Na
 8   No3
 9   TKN
10   Ot
```

```
Change any of the above ? Y/N...

Remove Selection No.
```

If the user wishes to output, say, the oxygen utilization rate for nitrification (O_n) instead of the TKN concentration, then a **9** would be entered at the prompt to designate the selected variable to be removed. The user is then prompted to enter the number of the replacement variable, **15** here, in the lower right window as shown below.

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)        g COD m-3
 2 Zba (autotrophs)     g COD m-3
 3 Ze (endog.)          g COD m-3
 4 Zi (prt unb COD)     g COD m-3
 5 Sads (adsorb.COD)    g COD m-3
 6 Senm (enmesh COD)    g COD m-3
 7 Nobp (prt bio N)     g N m-3
 8 Sbs (sol bio COD)    g COD m-3
 9 Na (ammonia N)       g N m-3
10 Nobs (sol org N)     g N m-3
11 No3 (nitrate N)      g N m-3
12 Alkalinity           mole m-3
13 Sus (sol unb COD)    g COD m-3
14 OUR carb        g O2 m-3 h-1
15 OUR nitr        g O2 m-3 h-1
16 OUR tot         g O2 m-3 h-1
17 Volatile solids      g VSS m-3
18 TKN                  g N m-3
```

```
SELECTED OUTPUTS

 1    Zbh
 2    Zba
 3    Sads
 4    Senm
 5    VSS
 6    Sbs
 7    Na
 8    No3
 9    TKN
10    Ot
```

```
Change any of the above ? Y/N...

Remove Selection No. 9

Replace with Parameter No.
```

The list of selected output variables is updated to reflect the change as shown below, and the user is prompted for more changes as shown below. These ten parameters will be listed either during computation of the diurnal response (**SIMULATE** option) or when the option **HARDCOPY OF RESULTS** is selected from the diurnal sub-menu.

```
POSSIBLE OUTPUT PARAMETERS

 1 Zbh (hetero.)        g COD m-3
 2 Zba (autotrophs)     g COD m-3
 3 Ze (endog.)          g COD m-3
 4 Zi (prt unb COD)     g COD m-3
 5 Sads (adsorb.COD)    g COD m-3
 6 Senm (enmesh COD)    g COD m-3
 7 Nobp (prt bio N)     g N m-3
 8 Sbs (sol bio COD)    g COD m-3
 9 Na (ammonia N)       g N m-3
10 Nobs (sol org N)     g N m-3
11 No3 (nitrate N)      g N m-3
12 Alkalinity           mole m-3
13 Sus (sol unb COD)    g COD m-3
14 OUR carb        g O2 m-3 h-1
15 OUR nitr        g O2 m-3 h-1
16 OUR tot         g O2 m-3 h-1
17 Volatile solids      g VSS m-3
18 TKN                  g N m-3
```

```
SELECTED OUTPUTS

 1    Zbh
 2    Zba
 3    Sads
 4    Senm
 5    VSS
 6    Sbs
 7    Na
 8    No3
 9    On
10    Ot
```

```
Change any of the above ? Y/N...
```

Once the desired set of ten output parameters has been selected and no more changes are required the user enters an **N** at the prompt in the lower right window. The diurnal sub-menu will re-appear as shown below.

```
          *****  SIMULATION OF DIURNAL RESPONSE  *****

        SIMULATE
        HARDCOPY OF RESULTS
        GRAPHICAL OUTPUT
        CHECK/CHANGE SCREEN/PRINTED OUTPUT
        CHECK/CHANGE INTEGRATION PARAMETERS
        STORE DATA ON DISK
        CHECK/CHANGE WASTAGE PATTERN

        RETURN TO MAIN MENU
```

```
        Press <Arrows> or <SpaceBar> to move selection

                Press <Return> to select option
```

Check/change integration parameters

When the **CHECK/CHANGE INTEGRATION PARAMETERS** option is selected from the diurnal menu the following list of three parameters appears on the screen.

```
**** INTEGRATION PARAMETERS ****

% Accuracy (0.01 to 1.0)                0.500
Theta (0.5 to 0.8)                      0.700
Data interval (10,15 or 30 mins)       15.000
                                             RETURN TO MENU
```

```
Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant
```

Default values for the three constants are provided in the program. These will appear in the list unless changes have already been made by the user during the current execution of the program. A short explanation of the function of the constants, and reasons for why the user may wish to alter the values, now follows.

% *Accuracy* and *Theta*
The first two parameters, % *Accuracy* and *Theta*, are utilized in the diurnal simulation by the integration algorithm in the automatic steplength adjustment procedure. % *Accuracy* relates to a limiting value for the integration error at each step. *Theta* is a safety factor limit the increase in steplength from one step to the next. Default values are provided in the program. These have been found suitable for most situations, and lead to acceptably short simulation times. It is likely that most users will have no need to adjust the values. However, under certain circumstances the integration algorithm may encounter difficulties when using the default values and adjustments must be made.

Problems with the integration algorithm usually manifest themselves in two ways. Firstly, a large number of cycles are required to converge to the solution in the **SIMULATE** option, well in excess of the usual two to five cycles. Secondly, in the graphical output certain parameters (typically soluble COD and oxygen utilization rate) may exhibit small oscillations ('noise' or 'spikyness') over a part of the diurnal cycle instead of a smooth change with time.

The problems usually occur in simulation of systems where there is a large difference in size between reactors or where there are very large changes in the influent flow. The solution normally lies in decreasing the parameter % *Accuracy* from the default of 0,5 to 0,1 or perhaps even smaller. Decreasing the parameter *Theta* from 9,7 to 9,6 or 9,5 can also improve performance of the integration algorithm. However, decreasing the parameters leads to longer simulation times as the integration is more rigorous.

Data interval

While simulating the response of a system it is necessary to store values of parameters at intervals over the diurnal cycle. This allows the user to print results and plot graphs of the response after the simulation has been completed.

The parameter *Data interval* specifies the time (in minutes) between values which are stored in memory. For example, with the default *Data interval* of 15 minutes there will be 97 discrete values of each parameter stored representing the response of the system over the cycle from 0h00 to 24h00.

The reason for reducing the *Data interval* to 10 minutes usually is to allow the response of rapidly-changing parameters to be tracked accurately. The penalty for reducing the *Data interval* is that the maximum allowable integration steplength may be constrained so as to store data at the correct times where otherwise a longer steplength may have been used. This will result in an increased simulation time.

> *The user should note that if the 'Data interval' is changed then the program requires that the* **SIMULATE** *option must be executed before attempting to print or plot data.*

Changes to any of the values can be made by the user. Any updated data will remain current during execution of the program, unless changed again. When execution is terminated and the program is re-run the original default values will appear.

Instructions for changing any of the data are provided in the window at the bottom of the screen. To update a value the highlighted block must be moved from **RETURN TO MENU** to the value which is to be changed. This is achieved by pressing the 'up arrow' (↑) or 'down arrow' (↓) keys or by pressing the spacebar (this scrolls the highlight from the top downwards). When the highlight appears on the value to be changed the user presses the **<RETURN>** key, and is prompted for a new value as shown below. On entering the new value the screen is updated and the highlight reverts to the **RETURN TO MENU** position.

```
              ****  INTEGRATION  PARAMETERS  ****

 % Accuracy (0.01 to 1.0)                 0.500
 Theta (0.5 to 0.8)                       0.700    New value =
 Data interval (10,15 or 30 mins)        15.000
                                                   RETURN TO MENU
```

```
      Hit <Arrows> or <SpaceBar> to move selection

   Hit <Return> to enter new value for selected constant
```

When the **RETURN TO MENU** option is selected the diurnal sub-menu re-appears with the highlight on the **SIMULATE** option as shown below.

```
***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU
```

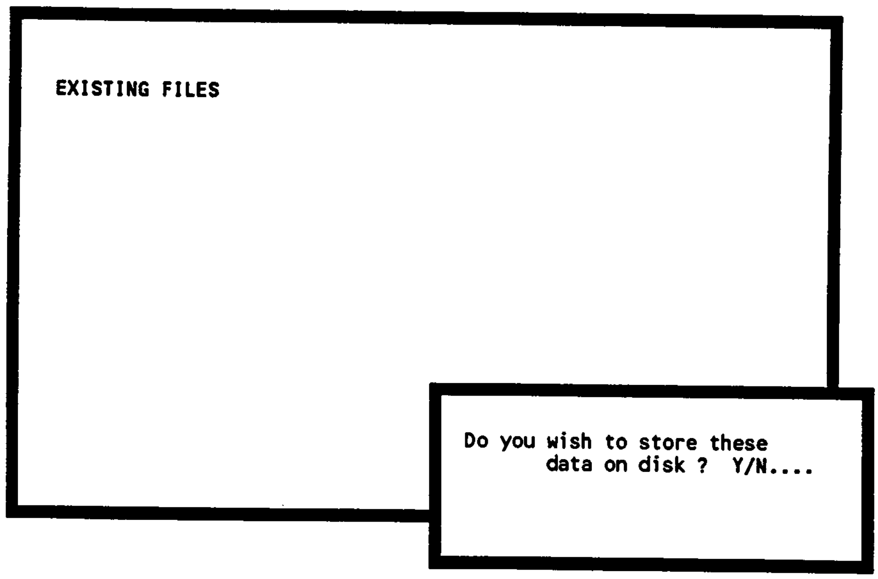
```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

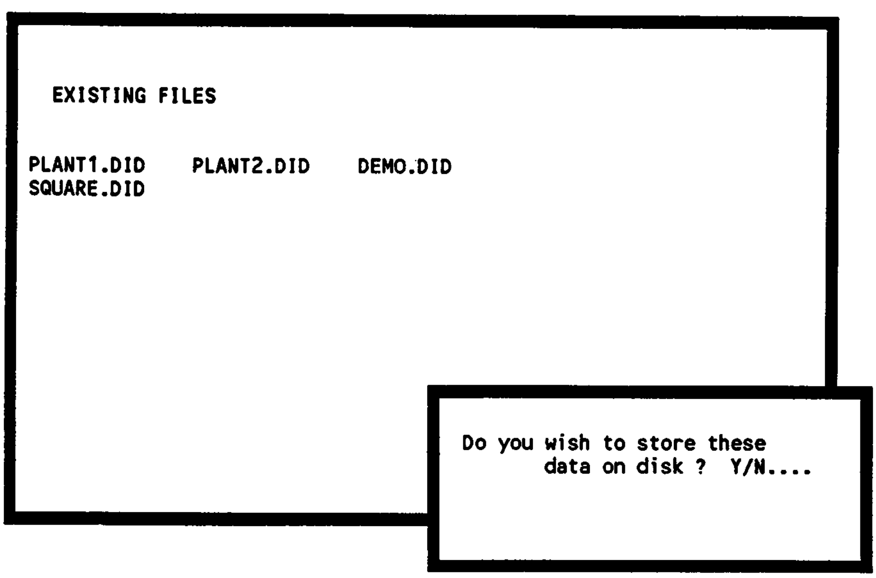
<u>Store data on disk</u>

The user may wish to store the results of a diurnal simulation on disk for off-line processing. For example, to allow a package such as *Lotus 1-2-3* access to the data.

When the option **STORE DATA ON DISK** is selected from the diurnal sub-menu for the first time two windows appear on the screen as shown below. The main window will not list any existing files as none have been created.



If the data have been stored on disk prior to selecting the **STORE DATA ON DISK** option then the names of all existing diurnal data files will be listed in three columns in the main window. The files are given the extension .DID. For example, the screen may appear as follows.



In the input window at the lower right the user enters a **Y** if the data are to be stored on disk. If an **N** is entered the program returns to the diurnal sub-menu. When a **Y** is entered at the prompt the user is requested to enter a file name in the input window as shown below. It is not necessary to type the extension .DID after the file name.

```
 EXISTING FILES

PLANT1.DID    PLANT2.DID    DEMO.DID
SQUARE.DID

                                   Enter data file name :

                                          e.g. PLANT1

                                          FileName =
```

The data are stored in a file of the specified name with the extension .DID on the disk (and in the directory) from where the program was executed. The user must wait for a few seconds while the data are being written to disk before continuing with program operation. During this delay the screen will appear as follows.

```
 EXISTING FILES

PLANT1.DID    PLANT2.DID    DEMO.DID
SQUARE.DID

                                   Enter data file name :

                                          e.g. PLANT1

                                          FileName = plant3
                                                Wait...
```

When the data have been written to disk the diurnal sub-menu re-appears with the highlight on the **SIMULATE** option as shown below.

```
***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

The data are stored in a Turbo Pascal *file of reals*. The size of the data file is different for **IAWPRC** or **UCTOLD** and is dependant on the number of reactors in the configuration and the *Data interval*. For example, with **UCTOLD** eighteen parameters are stored for each reactor. Therefore with a two reactor system simulated with a *Data interval* of 15 minutes (97 values for the 24-hour cycle), noting that 6 bytes are used to store each real value, the size of the data file will be

$$2 \cdot 18 \cdot 97 \cdot 6 = 20\,952 \text{ bytes.}$$

Details of the code used to write the data to disk are presented in **Appendix A**. This will enable the user to write a program to extract the data. The program **RETRIEVE** on the distribution disk can be used for creating a text file which can be imported into *Lotus 1-2-3* or Borland's *Quattro*.

Check/change wastage pattern

The **average** mixed liquor wastage rate required to establish the specified sludge age is calculated in the steady state section of the program. However, a number of situations may arise where the wastage rate at stages during the 24-hour cycle should not necessarily equal the average value. For example:

- In certain cases the user may wish to specify that wastage only takes place over a part of the cycle. A facility is included in the program to allow the wastage pattern over the 24-hour cycle to be manipulated. It is assumed that wastage is either on or off during each of 12 two-hour intervals. This corresponds to the division of the influent pattern. The wastage **rate** is the same over intervals when wastage occurs (and otherwise zero) and is such that the correct volume of mixed liquor is removed per day.

- Depending on the diurnal influent pattern situations may arise when the flow into the plant over one or more two-hour intervals is less than the average wastage rate. In this case the reactor volume will decrease and accordingly the wastage rate over such an interval should be reduced and the rate increased over another interval(s) where the influent flow is higher. In the program, before simulation of the diurnal response is started, the influent pattern is checked against the wastage rate. Adjustments are made to the waste pattern by switching wastage on or off during each of the 12 two-hour intervals in the cycle.

When the **CHECK/CHANGE WASTAGE PATTERN** option is selected from the diurnal sub-menu the screen will appear in a form similar to that shown below. This is the wastage pattern set up by the program and is in this case for the configuration used to illustrate program operation so far. That is, a two-reactor system with volumes of 2 and 5 m^3 operated at a sludge age of 5 days. The average wastage rate is $(2+5)/5 = 1{,}40$ m^3 d^{-1}. The influent flow follows a square wave pattern with flow for 12 hours between 06h00 and 18h00 and no flow for the remainder of the cycle.

****** SLUDGE WASTAGE PATTERN ******

Record No	Time (h)	Wastage On	Waste Flow (m3 d-1)
1	0.0	No	0.00
2	2.0	No	0.00
3	4.0	No	0.00
4	6.0	No	0.00
5	8.0	Yes	2.80
6	10.0	Yes	2.80
7	12.0	Yes	2.80
8	14.0	Yes	2.80
9	16.0	Yes	2.80
10	18.0	Yes	2.80
11	20.0	No	0.00
12	22.0	No	0.00

** Mean wastage rate =

1.40 m3/d

Switch any on/off? Y/N....

Because there is no influent flow for six of the 12 two-hour intervals the wastage must be switched off during these intervals. For the remaining six intervals the wastage rate must be set at twice the average value (i.e. 2,80 $m^3 d^{-1}$). The average appears at the lower left of the main window.

If the user wishes to make a change to the wastage pattern as defined in the main window a **Y** is entered at the prompt in the input window at the lower right of the screen. The user is then prompted to enter the number of the record where the wastage is to be changed from on to off or *vice versa* as shown below.

****** SLUDGE WASTAGE PATTERN ******

Record No	Time (h)	Wastage On	Waste Flow (m3 d-1)
1	0.0	No	0.00
2	2.0	No	0.00
3	4.0	No	0.00
4	6.0	No	0.00
5	8.0	Yes	2.80
6	10.0	Yes	2.80
7	12.0	Yes	2.80
8	14.0	Yes	2.80
9	16.0	Yes	2.80
10	18.0	Yes	2.80
11	20.0	No	0.00
12	22.0	No	0.00

** Mean wastage rate =

1.40 m3/d

Change in Record No.

When no more changes to the pattern are required the user enters an **N** at the prompt in the input window and the diurnal sub-menu re-appears with the highlight on the **SIMULATE** option as shown below.

***** SIMULATION OF DIURNAL RESPONSE *****

SIMULATE
HARDCOPY OF RESULTS
GRAPHICAL OUTPUT
CHECK/CHANGE SCREEN/PRINTED OUTPUT
CHECK/CHANGE INTEGRATION PARAMETERS
STORE DATA ON DISK
CHECK/CHANGE WASTAGE PATTERN

RETURN TO MAIN MENU

Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option

KINETIC CONSTANTS

When the **KINETIC CONSTANTS** option is selected from the main menu a sub-menu appears on the screen as shown below:

```
***** KINETIC CONSTANTS MENU *****

HETEROTROPHS
AUTOTROPHS
ARRHENIUS CONSTANTS

RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

The user may view and/or change the values of the kinetic constants for either the heterotrophs or the autotrophs, and the Arrhenius temperature dependency constants for certain of the kinetic constants. The appropriate set of constants is selected by moving the highlight to the desired option and pressing the **<RETURN>** key. For example, when the **HETEROTROPHS** option is selected when utilizing the **UCTOLD** model the screen will appear as shown below. ***The displayed data are the default values of the constants at 20° C.***

```
**** HETEROTROPHS ****

Mue max                    d-1                 3.200
Ks COD          (Ksh)  g COD m-3               5.000
Ks O2           (Koh)  g O2 m-3                0.002
B decay         (bh)   d-1                     0.620
Neta (growth)                                  0.330
Ks NO3          (Kno)  g N m-3                 0.100
Hydrolysis rate (Kmp)  d-1                     1.350
Ks hydrolysis   (Ksp)  g COD g-1 COD           0.027
Ammonification  (Kr)   m3 g-1 COD d-1          0.032
Ks NH3          (Kna)  g N m-3                 0.010
Adsorption rate (Ka)   g-1 COD m3 d-1          0.170
                                                    RETURN TO MENU
```

```
Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant
```

Default values for the kinetic constants and their temperature dependency constants are provided in the program. These will appear in the list unless changes have already been made by the user during the current execution of the program. The values are averages obtained in simulation of the observed response of a range of system configurations over a range of operating conditions.

Changes to any of the values can be made at this stage. Any updated data will remain current during execution of the program, unless changed again. When execution is terminated and the program is re-run the original default values will appear.

Instructions for changing any of the data are provided in the window at the bottom of the screen. To update a value the highlighted block must be moved from **RETURN TO MENU** to the value which is to be changed. This is achieved by pressing the 'up arrow' (↑) or 'down arrow' (↓) keys or by pressing the **spacebar** (this scrolls the highlight from the top downwards). When the highlight appears on the value to be changed the user presses the **<RETURN>** key, and is prompted for a new value as shown below. On entering the new value the screen is updated and the highlight reverts to the **RETURN TO MENU** position.

```
**** HETEROTROPHS ****

Mue max                   d-1                3.200
Ks COD             (Ksh)  g COD m-3          5.000
Ks O2              (Koh)  g O2 m-3           0.002
B decay            (bh)   d-1                0.620
Neta (growth)                                0.330
Ks NO3             (Kno)  g N m-3            0.100
Hydrolysis rate    (Kmp)  d-1                1.350    New value =
Ks hydrolysis      (Ksp)  g COD g-1 COD      0.027
Ammonification     (Kr)   m3 g-1 COD d-1     0.032
Ks NH3             (Kna)  g N m-3            0.010
Adsorption rate    (Ka)   g-1 COD m3 d-1     0.170
                                                      RETURN TO MENU
```

```
Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant
```

If the highlight is in the **RETURN TO MENU** position and the **RETURN** key is pressed the kinetic constants sub-menu will re-appear as shown below.

```
***** KINETIC CONSTANTS MENU *****

   HETEROTROPHS
   AUTOTROPHS
   ARRHENIUS CONSTANTS

   RETURN TO MAIN MENU
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

The user may either select one of these options or return to the main menu. When the user elects to return to the main menu the following window appears on the screen.

```
Do you want a hardcopy of kinetic constants? Y/N...
```

If a printer is attached to the computer the user may obtain a printout of all the kinetic and temperature dependency constants by entering a **Y**; else the **N** key is pressed. At a later stage, directly after simulating either the steady state or diurnal response, there is another opportunity to obtain a listing of the constants. After entering a **Y** or an **N** the program returns to the main menu with the highlight on the **STEADY STATE SIMULATION** option as shown below.

```
****** MAIN SIMULATION PROGRAM MENU ******

INFLUENT DATA
PLANT CONFIGURATION
OPERATING PARAMETERS
STEADY STATE SIMULATION
DIURNAL SIMULATION
KINETIC CONSTANTS
STOICHIOMETRY

EXIT FROM PROGRAM
```

```
Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option
```

STOICHIOMETRY

The user may view and/or change the values of the stoichiometric constants by selecting the **STOICHIOMETRY** option from the main menu and pressing the **<RETURN>** key. When utilizing the **UCTOLD** model the screen will appear as follows.

```
**** STOICHIOMETRIC PARAMETERS ****

Yield, hetero    (Yzh)    g COD g-1 COD  0.666
Frac inert       (Fe)     g COD g-1 COD  0.080
N in biomass     (Fzb,n)  g N g-1 COD    0.068
N in inert       (Fze,n)  g N g-1 COD    0.068
Yield, auto      (Yza)    g COD g-1 COD  0.150
COD:VSS ratio    (Fcv)    g COD g-1 VSS  1.480
Max adsorption   (Fma)    g COD g-1 COD  1.000
                                         RETURN TO MENU
```

```
Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant
```

Default values for the stoichiometric constants are provided in the program as is the case with the kinetic constants. These will be displayed unless changes have already been made by the user during the current execution of the program. The values are averages obtained in simulation of the observed response of a range of system configurations over a range of operating conditions.

Changes to any of the values can be made at this stage by following the procedure outlined in the window towards the base of the screen. Any updated data will remain current during execution of the program, unless changed again. When execution is terminated and the program is re-run the original default values will appear.

Instructions for changing any of the data are provided in the window at the bottom of the screen. To update a value the highlighted block must be moved from **RETURN TO MENU** to the value which is to be changed. This is achieved by pressing the 'up arrow' (↑) or 'down arrow' (↓) keys or by pressing the **spacebar** (this scrolls the highlight from the top downwards). When the highlight appears on the value to be changed the user presses the **<RETURN>** key, and is prompted for a new value as shown below. On entering the new value the screen is updated and the highlight reverts to the **RETURN TO MENU** position.

```
**** STOICHIOMETRIC PARAMETERS ****

Yield, hetero    (Yzh)   g COD g-1 COD  0.666
Frac inert       (Fe)    g COD g-1 COD  0.080
N in biomass     (Fzb,n) g N g-1  COD   0.068   New value =
N in inert       (Fze,n) g N g-1  COD   0.068
Yield, auto      (Yza)   g COD g-1 COD  0.150
COD:VSS ratio    (Fcv)   g COD g-1 VSS  1.480
Max adsorption   (Fma)   g COD g-1 COD  1.000
                                                RETURN TO MENU
```

```
Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant
```

When no more changes are required and the user elects to return to the main menu the following window appears on the screen.

```
**** STOICHIOMETRIC PARAMETERS ****

Yield, hetero    (Yzh)   g COD g-1 COD  0.666
Frac inert       (Fe)    g COD g-1 COD  0.080
N in biomass     (Fzb,n) g N g-1  COD   0.068
N in inert       (Fze,n) g N g-1  COD   0.068

Do you want a hardcopy of stoichiometric constants? Y/N...
```

```
Hit <Arrows> or <SpaceBar> to move selection

Hit <Return> to enter new value for selected constant
```

If a printer is attached to the computer the user may obtain a printout of the stoichiometric constants by entering a Y; else the N key is pressed. At a later stage, directly after simulating either the steady state or diurnal response, there is another opportunity to obtain a listing of the constants. After entering a Y or an N the program returns to the main menu with the highlight on the **STEADY STATE SIMULATION** option as shown overleaf.

****** MAIN SIMULATION PROGRAM MENU ******

INFLUENT DATA
PLANT CONFIGURATION
OPERATING PARAMETERS
STEADY STATE SIMULATION
DIURNAL SIMULATION
KINETIC CONSTANTS
STOICHIOMETRY

EXIT FROM PROGRAM

Press <Arrows> or <SpaceBar> to move selection

Press <Return> to select option